THE DISTRIBUTION OF THE THETA-FUNCTION SUM $S(p)$: LIMIT LAW, CHARACTER EXPANSION, SHARP TAILS AND EXTREME VALUES

SAMSON FRASER

ABSTRACT. For a prime p let $S(p)=\sum_{h,j=1}^{p-1}(-1)^{j+1+\lfloor hj/p\rfloor}$, a sum studied by Berndt, Bhat, Meyer, Xie and Zaharescu, and put $X_p=S(p)/p-\frac{1}{2}\log p$. We prove that X_p has a limiting distribution ν over the primes, and that the moments of X_p converge to those of ν. The law ν is symmetric about the constant $c_0=0.2399\ldots$. We express ν as an explicit series over odd Dirichlet characters, with coefficients given by $L(1,\chi)^2$ and an Euler product. This gives an exact formula for its variance, $\sigma^2\approx 0.39237$, and closed forms for the limiting means of X_p in residue classes; for example $c_0(3;\pm1)=c_0\pm\frac{\sqrt{3}\pi}{16}\prod_{\ell\equiv5\,(6)}(1+\frac{2}{(\ell-1)^2})$. The tails of ν are $\exp(-\exp((1+o(1))\sqrt{t/\kappa_H}))$ with $\kappa_H=2e^{\gamma}/\pi^2$. With the same constant, we show unconditionally and effectively that $X_p-c_0=\Omega_{\pm}((\log\log p)^2)$. Together these suggest $\limsup_p \pm(X_p-c_0)/(\log\log p)^2=\kappa_H$, of which we prove the lower bound. None of the results uses bounds for short Kloosterman sums. This is an AI-generated draft, not yet checked by a human expert; it relies on the asymptotic formula of an unrefereed companion paper.

1. INTRODUCTION

For an odd prime p, Berndt, Bhat, Meyer, Xie and Zaharescu [6] studied the sum

$$S(p)=\sum_{h=1}^{p-1}\sum_{j=1}^{p-1}(-1)^{j+1+\lfloor hj/p\rfloor},$$

which arises from a theta-function identity. They conjectured that $S(p)>0$, $S(p)>2p$ and $S(p)>3p$ beyond small thresholds, and that $C_1p<S(p)<C_2p^{1+\varepsilon}$ for large p. The companion paper [1] proves the asymptotic formula

$$S(p)=\frac{p}{2}\log p+c_0p+\frac{4p}{\pi}R_K(p)+O\Big(\frac{p}{\log p}\Big),\qquad R_K(n)=\sum_{\substack{3\le k<K\\ k\text{ odd}}}\frac{F_k(n)}{k^2},\tag{1}$$

for $\sqrt{p}(\log p)^4\le K\le p^{9/10}$, with $F_k(n)=\sum_{1\le t<k/2}\tan(\pi tn/k)$ and

$$c_0=\frac{K_0}{2}+\frac{\log 2}{6}+\frac{3}{\pi^2}\zeta'(2)=0.2399341036\ldots,\qquad K_0=\gamma+\log\frac{4}{\pi}$$

[1, Theorem B(ii)]. Its proof uses Weil's bound for Kloosterman sums, and nothing deeper. This paper studies the fluctuating term

$$X_p-c_0,\qquad X_p=\frac{S(p)}{p}-\frac{1}{2}\log p,$$

which by (1) is $\frac{4}{\pi}R_K(p)+o(1)$.

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Results. Let ν be the law of $c_0+\frac{4}{\pi}R_\infty$, where R_∞ is the L^2-limit of $R_K(x)$ for x Haar-distributed on $\Omega=\prod_{\ell\ \mathrm{odd}}\mathbb{Z}_\ell^\times$ (Section 3).

- *Limit law* (Theorem 1, Corollary 6.8). The empirical distribution of X_p, $p\le x$, converges to ν, together with all moments. The law ν is symmetric about c_0, and its support is unbounded in both directions. In residue classes the limits are the conditional laws, with means $c_0(q;r)$.
- *Character expansion* (Theorem 2). $R_\infty=\sum_\chi W(\chi)\chi$ over the odd primitive characters of odd conductor, with $W(\chi)$ explicit in $\tau(\chi)$, $L(1,\bar\chi)^2$ and an Euler product. Hence
$$\sigma^2=\mathrm{Var}(\nu)=\frac{1}{\pi^2}\sum_\chi\frac{|2-\chi(2)|^4|L(1,\chi)|^4|E(\chi)|^2}{f_\chi\varphi(f_\chi)^2}\approx 0.39237,$$
where $E(\chi)$ is the Euler product of Section 4, and $c_0(q;r)$ is a finite sum over the characters of conductor dividing q.
- *Extreme values* (Theorem 3). Unconditionally and effectively, for every $\varepsilon>0$ and large X, at least $X^{1-o(1)}$ primes $p\in(X,2X]$ satisfy $X_p-c_0\ge(\kappa_H-\varepsilon)(\log\log p)^2$, and as many satisfy $X_p-c_0\le-(\kappa_H-\varepsilon)(\log\log p)^2$, where $\kappa_H=2e^\gamma/\pi^2=0.3609\ldots$.
- *Sharp tails* (Theorem 4). $\nu((c_0+t,\infty))=\nu((-\infty,c_0-t))=\exp(-\exp((1+o(1))\sqrt{t/\kappa_H}))$, and $\log\mathbb{E}e^{\theta(Z-c_0)}\sim\kappa_H\theta(\log\theta)^2$ for $Z\sim\nu$.
- *Further consequences.* The class means for smooth moduli have a two-term asymptotic, $c_0(Q_K;\pm1)=c_0\pm(\kappa_H\log K(\log K-\gamma-\log 2+K_0)+C_Q+o(1))$ (Corollary 4.4). The density of primes with $\pm(X_p-c_0)>t$ is at least $\exp(-\exp((1+\varepsilon)\sqrt{t/\kappa_H}))$, uniformly up to $t\asymp(\log\log X)^2$ (Corollary 6.6).

The tails and the extreme values carry the same constant κ_H. Comparing them leads to a precise conjecture (Conjecture 5): $\limsup\pm(X_p-c_0)/(\log\log p)^2=\kappa_H$. Theorem 3 proves $\limsup\ge\kappa_H$ and $\liminf\le-\kappa_H$; the reverse inequalities are open. It is the analogue of the Montgomery–Vaughan conjecture for $L(1,\chi)$, and Theorem 4 is the analogue of Granville and Soundararajan's theorem on the distribution of $L(1,\chi)$.

Earlier limit laws for cotangent and Dedekind sums [5, 2] average over all fractions a/q. By contrast, $S(p)$ involves one point per modulus. The numerical values of $c_0(q;r)$ were found in [1] by summing over $d\le 8001$ and extrapolating; Theorem 2 gives them in closed form.

What is used. Standard references for the analytic tools are [3, 4]. Besides (1), we use the elementary lemmas of [1, Sections 4 and 9], recalled in Section 2. We also use the prime number theorem in progressions (Siegel–Walfisz) for Theorem 1, and the Bombieri–Vinogradov theorem for the moments. Theorem 3 uses Gallagher's prime number theorem and the Landau–Page lemma, and it avoids Siegel's theorem. No bounds for short Kloosterman sums are needed. In particular, none of the results depends on the Bourgain–Garaev bound, on which the lower half of the conjecture $S(p)>C_1p$ rests in [1].

The numbers quoted come from exact values of $S(p)$ for all primes $p\le 5.5\cdot10^6$, from Haar samples of the model, and from the code `code/char_expansion.py`, all in the repository that accompanies the main paper [1].

2. Notation and inputs

Results of [1] are cited by their numbers there; in particular (9.1) and (9.2) are equations of [1]. For odd $d\ge3$ put
$$G_d(n)=\sum_{\substack{1\le j<d/2\\(j,d)=1}}\tan\frac{\pi jn}{d},$$

an odd function of $n \bmod d$. We use the following results of [1].

- *Theorem B(ii)* is (1). In particular, for $p \in (X, 2X]$ and $K_2 = \sqrt{2X}(\log 2X)^4$ we have $X_p - c_0 = \frac{4}{\pi} R_{K_2}(p) + O(1/\log X)$.
- *Corollary 4.3.* (a) For odd $m \geq 1$, $\frac{m}{\pi}(\log m + K_0) - \frac{7\pi}{72m} \leq F_m(1) \leq \frac{m}{\pi}(\log m + K_0)$. (b) For odd $k \geq 3$ and every integer r, $|F_k(r)| \leq F_k(1) \leq \frac{k}{\pi}(\log k + K_0)$, and $F_k(-1) = -F_k(1)$.
- *Lemma 9.1.* $F_k(n) = \sum_{d|k,\, d\geq 3} G_d(n)$ for odd k. Hence $R_K = \sum_{3\leq d<K} w_d(K) G_d/d^2$, with $w_d(K) = \sum_{m \text{ odd},\, dm<K} m^{-2} \in [1, \pi^2/8]$. Also $R_{K_2} - R_{K_1} = \sum_d w_d(K_1, K_2) G_d/d^2$, with $0 \leq w_d(K_1, K_2) \leq \min(\pi^2/8, 2d/K_1)$.
- *Lemma 9.2.* Let $g \mid d$. (a) $A_{d,g}(s) := \frac{g}{d} \sum_{i<d/g} G_d(s + gi) = \sum_{j<d/2,(j,d)=1} \tan(\pi j s/g)$. (b) $|A_{d,g}(s)| \leq \frac{\tau(d)}{\pi} g(\log g + 2)$ for $g \geq 3$; in particular $|G_d(n)| \leq \frac{\tau(d)}{\pi} d(\log d + 2)$. (c) Let $A^*_{d,g}(s)$ be the average of $G_d(x)$ over the reduced residues $x \bmod d$ with $x \equiv s \pmod g$, $(s, g) = 1$. Then $A^*_{d,1} = 0$, and $|A^*_{d,g}(s)| \leq C\tau(d)^2 \lambda(d) g$ for $g \geq 3$, where $\lambda(n) = (\log\log 3n)(\log n + 2)$.
- *The observation before (9.2).* If x is uniform on the reduced residues modulo k with $x \equiv r \pmod{(k, q)}$, then for $d \mid k$ the residue $x \bmod d$ is uniform on the reduced residues modulo d with $x \equiv r \pmod{(d, q)}$.
- *The tangent identity* (proof of Lemma 9.2(a)). For odd n and real y avoiding poles, $\sum_{b<n} \tan(y + \pi b/n) = n \tan ny$.
- *(9.1).* If L is a common multiple of d, d' and x is uniform on $(\mathbb{Z}/L\mathbb{Z})^\times$, then $\mathbb{E}[G_d(x) G_{d'}(x)] = \mathbb{E}_s[A^*_{d,g}(s) A^*_{d',g}(s)]$ with $g = (d, d')$ and s uniform on $(\mathbb{Z}/g\mathbb{Z})^\times$. In particular the expectation vanishes when $g = 1$.
- *Lemmas 9.3(b) and 9.4.* $\sum_{X<n\leq 2X} (R_{K_2}(n) - R_Y(n))^2 \ll X(\log X)^{14}/Y$ for $3 \leq Y < K_2$.
- *Theorems E and F.* Theorem E(i) says that $|X_p| \ll_A (\log\log p)^2$ for all $p \in (X, 2X]$ outside a set of size $O_A(X(\log X)^{-A})$. Theorem E(ii) says that $\sum_{X<p\leq 2X} (X_p - c_0)^2 \ll X/\log X$, with an ineffective constant. Theorem F gives the limiting mean $c_0(q; r)$ of X_p over $p \equiv r \pmod q$, with $c_0(q; r) = c_0 + \frac{\pi}{2} \sum_d A^*_{d,(d,q)}(r)/d^2$ by (9.2).

3. The limiting distribution

Let $\Omega = \prod_{\ell \text{ odd prime}} \mathbb{Z}_\ell^\times$ with its Haar probability measure $\mathbb{P}$. For odd k and $x \in \Omega$, $x \bmod k$ is a reduced residue, and we put $F_k(x) = F_k(n)$ and $G_d(x) = G_d(n)$ for any integer $n \equiv x \pmod k$, resp. $\pmod d$. For every odd Q, $x \bmod Q$ is uniformly distributed over $(\mathbb{Z}/Q\mathbb{Z})^\times$. Hence expectations of functions of $x \bmod Q$ are the averages of [1, Section 9], and (9.1) holds under $\mathbb{P}$. As in [1], $R_K(x) = \sum_{3\leq k<K,\ k \text{ odd}} F_k(x)/k^2$. For a prime p put

$$X_p = \frac{S(p)}{p} - \frac{1}{2}\log p.$$

Lemma 3.1. *For $K \geq 3$ let*

$$\mathcal{E}(K) = \sup_{K'>K} \mathbb{E}\big[(R_{K'} - R_K)^2\big].$$

Then $\mathcal{E}(K) \to 0$ as $K \to \infty$. Hence R_K converges in $L^2(\Omega)$ to a limit R_∞, with $\mathbb{E}[R_\infty] = 0$ and

$$\mathbb{E}[R_\infty^2] = \frac{\pi^4}{64} \sum_{d,d'\geq 3 \text{ odd}} \frac{\mathbb{E}[G_d G_{d'}]}{(dd')^2},$$

the series being absolutely convergent.

Proof. By Lemma 9.1, $R_{K'} - R_K = \sum_{d<K'} w_d(K,K')G_d/d^2$ with $0 \le w_d(K,K') \le \min(\pi^2/8, 2d/K)$. By (9.1) and Lemma 9.2(c), $\mathbb{E}[G_dG_{d'}] = 0$ unless $g = (d,d') \ge 3$. In that case

$$|\mathbb{E}[G_dG_{d'}]| \le C\tau(d)^2\tau(d')^2\lambda(d)\lambda(d')g^2, \qquad \lambda(n) = (\log\log 3n)(\log n + 2).$$

Write $d = ga$ and $d' = ga'$, and drop the coprimality of a, a'; this gives the upper bound

$$\mathbb{E}[(R_{K'} - R_K)^2] \le C\sum_{g\ge 3}\sum_{a,a'\ge 1} \min\Big(1, \frac{2ga}{K}\Big)\min\Big(1, \frac{2ga'}{K}\Big)\frac{\tau(ga)^2\tau(ga')^2\lambda(ga)\lambda(ga')}{g^2a^2a'^2}.$$

Since $\tau(ga) \le \tau(g)\tau(a)$ and $\lambda(n) \ll n^{1/10}$, the summand is at most $\ll g^{-2+1/2}(aa')^{-2+1/2}$ uniformly in K, and this majorant is summable. Each summand tends to 0 as $K \to \infty$, so $\mathcal{E}(K) \to 0$ by dominated convergence. The formula for $\mathbb{E}[R_\infty^2]$ follows because $w_d(K) \to \pi^2/8$ for every d, and the same majorant applies. □

Lemma 3.2. *Let $\mathcal{P}_X$ be the set of primes in $(X, 2X]$, and fix $K \ge 3$. Then*

$$\limsup_{X\to\infty} \frac{1}{\#\mathcal{P}_X}\sum_{p\in\mathcal{P}_X}\Big(X_p - c_0 - \frac{4}{\pi}R_K(p)\Big)^2 \le \frac{16}{\pi^2}\mathcal{E}(K).$$

Proof. Let $K_2 = \sqrt{2X}(\log 2X)^4$ and $Y = (\log X)^{16}$. By Theorem B(ii), $X_p - c_0 = \frac{4}{\pi}R_{K_2}(p) + O(1/\log X)$ for $p \in \mathcal{P}_X$. We have

$$R_{K_2} - R_K = (R_{K_2} - R_Y) + (R_Y - R_K).$$

- *The first part.* By Lemmas 9.3(b) and 9.4, exactly as in the proof of Theorem E, $\sum_{X<n\le 2X}(R_{K_2}(n) - R_Y(n))^2 \ll X(\log X)^{-2}$. So its mean square over $\mathcal{P}_X$ is $O(1/\log X)$.
- *The second part.* Expand $(R_Y - R_K)^2$ by Lemma 9.1 into terms $w_dw_{d'}G_d(p)G_{d'}(p)/(dd')^2$ with $d, d' < Y$. Each $G_d(p)G_{d'}(p)$ depends only on $p \bmod \operatorname{lcm}(d,d') \le Y^2 = (\log X)^{32}$. So, as in the proof of Theorem E(ii), the Siegel–Walfisz theorem gives

$$\frac{1}{\#\mathcal{P}_X}\sum_{p\in\mathcal{P}_X}(R_Y(p) - R_K(p))^2 = \mathbb{E}[(R_Y - R_K)^2] + o(1) \le \mathcal{E}(K) + o(1).$$

  Here the Siegel–Walfisz error, after division by $\#\mathcal{P}_X$, is $O(Y^4(\log X)\exp(-c\sqrt{\log X})) = o(1)$.

Minkowski's inequality combines the three pieces. □

Theorem 1. *Let ν be the distribution of $c_0 + \frac{4}{\pi}R_\infty$ under $\mathbb{P}$.*

(a) *As $x \to \infty$,*

$$\frac{1}{\pi(x)}\sum_{p\le x} f(X_p) \to \int f\,d\nu$$

for every continuous $f : \mathbb{R} \to \mathbb{R}$ with $|f(t)| \le C_f(1+t^2)$. In particular,

$$\frac{1}{\pi(x)}\sum_{p\le x}(X_p - c_0)^2 \to \sigma^2 := \frac{16}{\pi^2}\mathbb{E}[R_\infty^2] = \frac{\pi^2}{4}\sum_{d,d'}\frac{\mathbb{E}[G_dG_{d'}]}{(dd')^2}.$$

(b) *The support of ν is unbounded above and below. Hence, for every $M > 0$, both sets*

$$\Big\{p :\ S(p) > \frac{p}{2}\log p + Mp\Big\}, \qquad \Big\{p :\ S(p) < \frac{p}{2}\log p - Mp\Big\}$$

have positive lower density in the primes.

(c) *More generally, let $q \geq 1$ and $(r,q)=1$, and write $q = 2^e q'$ with q' odd. Then the X_p with $p \equiv r \pmod q$ have as limiting distribution the law of $c_0 + \frac{4}{\pi}R_\infty$ conditioned on $x \equiv r \pmod{q'}$. Its mean is the constant $c_0(q;r)$ of Theorem F. (Only q' matters, because R_K involves only odd moduli and $(d,q)=(d,q')$ for odd d; in particular $c_0(q;r)=c_0$ when q is a power of 2.)*

Proof. (a) Let f be bounded with Lipschitz constant at most 1. Then

$$\Big|\frac{1}{\#\mathcal{P}_X}\sum_{p\in\mathcal{P}_X} f(X_p) - \mathbb{E}f\Big(c_0+\frac{4}{\pi}R_\infty\Big)\Big| \leq \Delta_1+\Delta_2+\Delta_3,$$

where:

- Δ_1 is the root mean square of $X_p - c_0 - \frac{4}{\pi}R_K(p)$ over $\mathcal{P}_X$; by Lemma 3.2 it is at most $\frac{4}{\pi}\mathcal{E}(K)^{1/2}+o(1)$.
- Δ_2 is the difference between the averages of $f(c_0+\frac{4}{\pi}R_K(p))$ over $\mathcal{P}_X$ and of $f(c_0+\frac{4}{\pi}R_K(x))$ over Ω. Since $R_K(p)$ depends only on p modulo the fixed number $\operatorname{lcm}\{k<K\}$, the prime number theorem for arithmetic progressions gives $\Delta_2 \to 0$.
- $\Delta_3 = \frac{4}{\pi}\|R_K - R_\infty\|_2 \leq \frac{4}{\pi}\mathcal{E}(K)^{1/2}$.

Letting $X\to\infty$ and then $K\to\infty$ proves the convergence over each dyadic block, and therefore over $p\leq x$. Bounded Lipschitz functions determine weak convergence. For the second moments, Minkowski's inequality in $L^2(\mathcal{P}_X)$ gives

$$\big|\,\|X_p - c_0\| - \|\tfrac{4}{\pi}R_K(p)\|\,\big| \leq \tfrac{4}{\pi}\mathcal{E}(K)^{1/2}+o(1).$$

Also $\|R_K(p)\|_{L^2(\mathcal{P}_X)} \to \|R_K\|_{L^2(\Omega)}$ by the prime number theorem modulo the fixed number $\operatorname{lcm}\{k<K\}$, and $\big|\|R_K\|_2 - \|R_\infty\|_2\big| \leq \mathcal{E}(K)^{1/2}$. Hence the second moments converge to σ^2. Together with weak convergence this gives uniform integrability of $(X_p-c_0)^2$, and hence the statement for $|f| \leq C_f(1+t^2)$.

(b) Fix an odd Q and $r\in\{1,-1\}$, and let $\mathcal{C} = \{x \equiv r \pmod Q\}$, so that $\mathbb{P}(\mathcal{C}) = 1/\varphi(Q)$. Conditionally on $\mathcal{C}$, $x \bmod d$ is uniform over the reduced residues $\equiv r \pmod{(d,Q)}$, as in the observation before (9.2). Hence $\mathbb{E}[G_d \mid \mathcal{C}] = A^*_{d,(d,Q)}(r)$. Lemma 9.2(c) and $(d,Q)\leq Q$ make $\sum_d |A^*_{d,(d,Q)}(r)|/d^2$ converge. Together with L^2 convergence and Lemma 9.1 (whose weights $w_d(K)$ increase to $\pi^2/8$) this gives

$$\mathbb{E}[R_\infty \mid \mathcal{C}] = \frac{\pi^2}{8}\sum_{d\geq 3 \text{ odd}} \frac{A^*_{d,(d,Q)}(r)}{d^2} = \frac{\pi^2}{8}\Big(r\sum_{d\mid Q,\, d\geq 3}\frac{G_d(1)}{d^2} + \sum_{d\nmid Q}\frac{A^*_{d,(d,Q)}(r)}{d^2}\Big), \tag{2}$$

where for $d\mid Q$ we used $A^*_{d,d}(r) = G_d(r) = rG_d(1)$ (G_d is odd).

Every term of $G_d(1) = \sum_{j<d/2,(j,d)=1}\tan(\pi j/d)$ is positive, and $G_\ell(1) = F_\ell(1) \geq \frac{\ell}{\pi}(\log\ell + K_0) - \frac{7\pi}{72\ell}$ for primes ℓ (Corollary 4.3(a)).

Now take $K\geq 5$ and $Q = Q_K = \prod_{3\leq\ell<K}\ell^{a_\ell}$, with a_ℓ minimal such that $\ell^{a_\ell} > K^3$. The first sum in (2) is at least $\sum_{3\leq\ell<K} G_\ell(1)/\ell^2 \geq \frac{1}{\pi}\sum_{3\leq\ell<K}\frac{\log\ell}{\ell} - O(1) \to \infty$.

We show the second sum tends to 0. Write $d = gh$ with $g=(d,Q)$ and $h\geq 2$; then $\tau(d)\leq \tau(g)\tau(h)$ and $\lambda(d)\leq\lambda'(g)\lambda'(h)$, where $\lambda'(n) = (1+\log\log 3n)(\log n+2)$. By Lemma 9.2(c) the term for d is $\ll \tau(g)^2\tau(h)^2\lambda'(g)\lambda'(h)/(gh^2)$. Let $\eta_K = 1/\log K$.

- Since $\lambda'(g) \ll (\log K)^3 g^{\eta_K}$, for K-smooth g

$$\sum_{g\ K\text{-smooth}}\frac{\tau(g)^2\lambda'(g)}{g} \ll (\log K)^3\prod_{\ell<K}\Big(\sum_{b\geq 0}(b+1)^2\ell^{-b(1-\eta_K)}\Big) \ll (\log K)^{3+4e},$$

because $\ell_K^{\eta} \le e$ for $\ell < K$.

- If h has a prime factor $\ge K$, then $\sum_h \tau(h)^2 \lambda'(h) h^{-2} \ll K^{-1/2}$, so these d contribute $\ll (\log K)^{3+4e} K^{-1/2}$.
- Otherwise $v_\ell(d) > a_\ell$ for some $\ell < K$. Then $\ell^{a_\ell} \mid g$ and $\ell \mid h$, and these d contribute $\ll \sum_{\ell<K} (a_\ell+1)^2 \lambda'(\ell^{a_\ell}) (\log K)^{3+4e} \ell^{-a_\ell} \ll (\log K)^{O(1)} K^{-2}$.

Hence $\mathbb{E}[R_\infty \mid x \equiv \pm 1 \pmod{Q_K}] \to \pm\infty$ as $K \to \infty$. A conditional expectation over a set of positive measure lies between the essential infimum and the essential supremum, so R_∞ is essentially unbounded above and below, and so is the support of ν. Finally, $\nu((M,\infty)) > 0$ and $\nu((-\infty,-M)) > 0$, and the portmanteau theorem applied to these open sets gives the positive lower densities.

(c) Same proof as (a), with the Siegel–Walfisz theorem and the prime number theorem applied in the class $r \bmod q$, and $\mathbb{P}$ replaced by $\mathbb{P}(\cdot \mid x \equiv r \bmod q')$. The mean is computed as in (b), and equals $c_0(q;r)$ by (9.2). □

Corollary 3.3. *The error term in the asymptotic formula of* [1, Theorem C] *cannot be replaced by $o(p)$, even on a set of primes of density* 1. *More precisely, for every M, a positive proportion of primes satisfy $|S(p) - \frac{p}{2}\log p - c_0 p| > Mp$. Also, Theorem E(ii) holds as an asymptotic:*

$$\sum_{X<p\le 2X} \Big(\frac{S(p)}{p} - \frac{1}{2}\log p - c_0\Big)^2 \sim \sigma^2(\pi(2X) - \pi(X)).$$

Remark 3.4 (Symmetry). The law ν is symmetric about c_0. Indeed $x \mapsto -x$ preserves $\mathbb{P}$, and $R_K(-x) = -R_K(x)$ because every F_k is odd, so $R_\infty(-x) = -R_\infty(x)$. Hence all odd moments of ν about c_0 vanish, and by Corollary 6.8 below, $\frac{1}{\pi(x)}\sum_{p\le x}(X_p - c_0)^{2j+1} \to 0$ for every $j \ge 0$. The observed skewness of $X_p - c_0$ over the primes in $(10^6, 5.5\cdot 10^6]$ is -0.0006.

Remark 3.5 (Numerical value of σ). The truncated variance $\sigma_K^2 = \frac{16}{\pi^2}\mathbb{E}[R_K^2]$ is a finite sum over pairs $k,k' < K$, by $\mathbb{E}[F_k F_{k'}] = \mathbb{E}_s[\langle F_k\rangle_{g,s}\langle F_{k'}\rangle_{g,s}]$ with $g = (k,k')$, where $\langle F_k\rangle_{g,s}$ is the average of F_k over the reduced residues $\equiv s \pmod g$, as in [1, Theorem F]. The program `code/haar_variance.c` computes it in double precision, without sampling. It gives

$$\sigma_{2001} = 0.61974, \qquad \sigma_{8001} = 0.62420, \qquad \sigma_{16001} = 0.62515, \qquad \sigma_{32001} = 0.62569.$$

The increments satisfy $K^2\,\Delta\mathbb{E}[R_K^2] \approx c\log K$ with $c \approx 3$, so the tail beyond K is about $(c\log K + c')/K$. Fits of $\mathbb{E}[R_K^2] = \mathbb{E}[R_\infty^2] - P(\log K)/K$, with P of degree 1 to 3, to K between 2001 and 32001 leave residuals below 10^{-5}. They give

$$\sigma^2 = 0.3924 \pm 0.0001, \qquad \sigma = 0.6264 \pm 0.0001.$$

(The quoted uncertainty is the spread over the fits.) Theorem 2(b) below gives the sharper value $\sigma^2 = 0.392369 \pm 0.000002$ (Remark 4.3). The tempting closed form $\sigma^2 = \pi/8 = 0.39270$ is not supported by the data: it would need a tail about 35% larger than the observed trend.

The observed standard deviations of X_p are 0.6237 for the 17816 primes in $(10^3, 2\cdot 10^5]$ and 0.6268 for the 60514 primes in $(2\cdot 10^5, 10^6]$ (our computation; it reproduces [1, Table 3]). [1, Table 3] gives 0.624–0.628 for the ranges up to $6\cdot 10^6$. The Gaussian sampling errors $\sigma/\sqrt{2n}$ are 0.003 and 0.002; they understate the true error somewhat for a law with unbounded support, but the agreement is clear. Theorem 1 thus explains why the standard deviation in [1, Table 3] does not change with p.

The whole distribution agrees, not only the variance. The program `code/haar_sample.c` draws $2\cdot 10^5$ samples of $\frac{4}{\pi}R_{4001}(x)$ with x Haar-random. (See also Figure 1.) Their Kolmogorov–Smirnov distance from the empirical distribution of $X_p - c_0$ over the 60514 primes in $(2\cdot 10^5, 10^6]$ is 0.0043, below the 5% critical value 0.0063. The quantiles agree to within 0.025

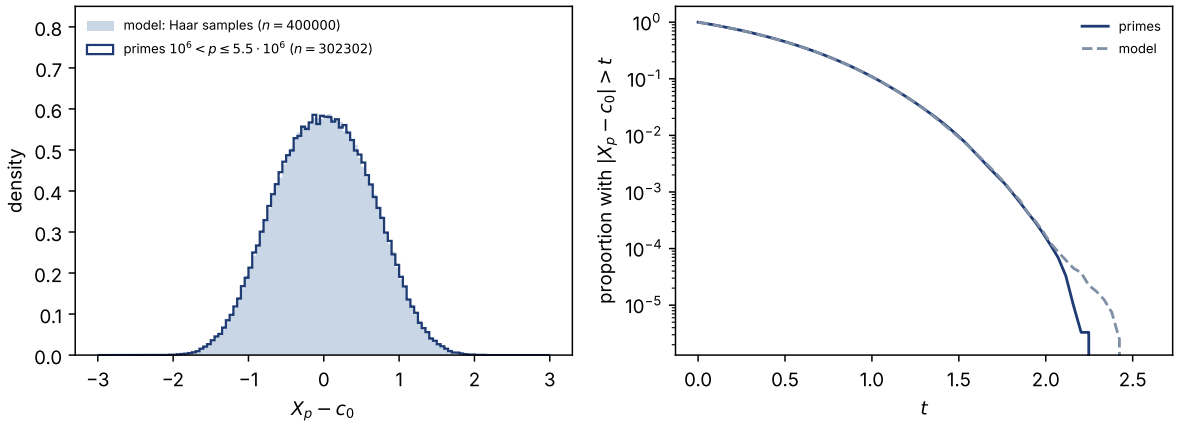


FIGURE 1. Left: the distribution of $X_p - c_0$ over the 302302 primes in $(10^6, 5.5 \cdot 10^6]$, against $4 \cdot 10^5$ Haar samples of $\frac{4}{\pi} R_{8001}$. Right: the proportion with $|X_p - c_0| > t$, on a log scale. The curves separate only near the sampling floor $1/n$.

from the 0.1% to the 99.9% quantile (`data/limit_law_comparison.txt`; the variance data are in `data/haar_variance_K32001.txt`).

Remark 3.6 (What is new here). Part (a) is the standard argument for limit-periodic functions in B^2: it uses Theorem B(ii) and Lemma 9.3, with the Siegel–Walfisz theorem to pass to primes. The density statements in (b) also follow, without (a), from Theorems E(ii) and F and the Cauchy–Schwarz inequality, once one knows that

$$\sup_q c_0(q;1) = +\infty \qquad \text{and} \qquad \inf_q c_0(q;-1) = -\infty.$$

That is the computation (2) at $Q = Q_K$, and it is the new input. In [1] the constants $c_0(q;r)$ appear only numerically [1, Table 2]. We know of no earlier limiting-distribution result for $S(p)$. The limit laws of Vardi [5] for Dedekind sums and of Bettin and Drappeau [2] for cotangent sums average over all fractions a/q, whereas $S(p)$ involves one point per modulus.

4. The character expansion of the limit law

The limit R_∞ has an explicit expansion in Dirichlet characters. It gives exact formulas for σ^2 and for the constants $c_0(q;r)$ of Theorem F, which [1] computed only numerically.

Let $\mathcal{X}$ be the set of odd primitive Dirichlet characters χ whose conductor $f = f_\chi$ is odd (so $f \ge 3$). Every such χ defines a continuous character $x \mapsto \chi(x \bmod f)$ of Ω. For $\chi \in \mathcal{X}$ let $\tau(\chi)$ be its Gauss sum, and put

$$E(\chi) = \prod_{\ell \mid f} \Big(1 - \frac{1}{\ell^2}\Big)^{-1} \prod_{\ell \nmid 2f} \Big(1 + \frac{(1-\chi(\ell))(1-\bar\chi(\ell)/\ell)}{(1-1/\ell)(\ell^2-1)}\Big),$$

$$W(\chi) = \frac{i\,\tau(\chi)\chi(2)}{4f\varphi(f)}\,(2-\bar\chi(2))^2\,L(1,\bar\chi)^2\,E(\chi),$$

the products running over primes ℓ and converging absolutely.

Theorem 2. (a) *In* $L^2(\Omega)$, $R_\infty = \sum_{\chi \in \mathcal{X}} W(\chi)\,\chi$.

(b) *Consequently*

$$\sigma^2 = \frac{16}{\pi^2}\sum_{\chi\in\mathcal{X}}|W(\chi)|^2 = \frac{1}{\pi^2}\sum_{\chi\in\mathcal{X}}\frac{|2-\chi(2)|^4\,|L(1,\chi)|^4\,|E(\chi)|^2}{f\,\varphi(f)^2}.$$

(c) *For $q\geq 1$ with odd part q' and $(r,q)=1$, the constant of Theorem F is the finite sum*

$$c_0(q;r) = c_0 + \frac{4}{\pi}\sum_{\chi\in\mathcal{X},\ f_\chi\mid q'} W(\chi)\chi(r).$$

For example, $\mathcal{X}$ contains a single character of conductor 3, namely χ_{-3}, and

$$c_0(3;\pm 1) = c_0 \pm \frac{\sqrt{3}\,\pi}{18}\cdot\frac{9}{8}\prod_{\ell\equiv 5\ (6)}\Big(1+\frac{2}{(\ell-1)^2}\Big) = c_0 \pm 0.398484\ldots,$$

so $c_0(3;1) = 0.638418\ldots$ and $c_0(3;2) = -0.158550\ldots$.

The proof rests on two lemmas.

Lemma 4.1. *Let $\chi\in\mathcal{X}$ have conductor f. Then*

$$h(\chi) := \sum_{0<a<f/2}\chi(a) = \frac{\tau(\chi)}{\pi i}(2-\bar\chi(2))L(1,\bar\chi),$$

$$t(\chi) := \sum_{a \bmod f}\bar\chi(a)\tan\frac{\pi a}{f} = -\frac{2f}{\pi}(2\chi(2)-1)L(1,\bar\chi).$$

Proof. For $0<x<1$, $x\neq\frac12$, the square wave has the symmetrically convergent Fourier series $\mathbf{1}_{(0,1/2)}(x)-\frac12 = \sum_{n\text{ odd}}\frac{e(nx)}{\pi i n}$. Put $x=a/f$, multiply by $\chi(a)$ and sum over a. Since $\sum_a\chi(a)=0$ and $\sum_a\chi(a)e(na/f)=\bar\chi(n)\tau(\chi)$ (because χ is primitive), we get $h(\chi)=\frac{\tau(\chi)}{\pi i}\sum_{n\text{ odd}}\frac{\bar\chi(n)}{n}$. As $\bar\chi$ is odd, this sum is $2\sum_{n>0\text{ odd}}\bar\chi(n)/n = (2-\bar\chi(2))L(1,\bar\chi)$.

For t, use $\pi\tan\pi x = -\sum_{n\in\mathbb{Z}}\frac{1}{x+\frac12+n}$ (summed symmetrically), so that $\pi\tan\frac{\pi a}{f} = -\sum_n\frac{2f}{2a+f(2n+1)}$. As a runs modulo f and n over $\mathbb{Z}$, $k=2a+f(2n+1)$ runs once over the odd integers, with $a\equiv k\bar2 \pmod f$, so $\bar\chi(a)=\chi(2)\bar\chi(k)$. Hence $t(\chi) = -\frac{2f}{\pi}\chi(2)\sum_{k\text{ odd}}\frac{\bar\chi(k)}{k} = -\frac{2f}{\pi}\chi(2)(2-\bar\chi(2))L(1,\bar\chi)$, which is the stated formula. (In both computations, the symmetric truncations of the double sum and of the sum over k differ by $O(1/N)$.) □

Lemma 4.2. *Let $d\geq 3$ be odd and let χ be a character modulo d. If χ is even, then $\sum_x G_d(x)\bar\chi(x)=0$. If χ is odd and induced by $\chi^*\in\mathcal{X}$ of conductor $f\mid d$, let m' be the product of the primes dividing d but not f. Then*

$$\frac{1}{\varphi(d)}\sum_{x\in(\mathbb{Z}/d\mathbb{Z})^\times}G_d(x)\bar\chi(x) = \frac{d}{f\varphi(d)}\,h(\chi^*)\,t(\chi^*)\prod_{\ell\mid m'}(1-\chi^*(\ell))\Big(1-\frac{\bar\chi^*(\ell)}{\ell}\Big).$$

Proof. G_d is odd, which gives the first claim. For the second, substitute $u=jx$ in $\sum_x\bar\chi(x)\sum_j\tan(\pi jx/d)$. The left side becomes $\frac{1}{\varphi(d)}H\cdot T$, where

$$H = \sum_{\substack{0<j<d/2\\(j,d)=1}}\chi^*(j), \qquad T = \sum_{u\in(\mathbb{Z}/d\mathbb{Z})^\times}\bar\chi^*(u)\tan\frac{\pi u}{d}.$$

Remove the condition $(\cdot,m')=1$ by Möbius inversion over $e\mid m'$.

- In H, the inner sums are $\chi^*(e)\sum_{0<i<d/(2e)}\chi^*(i)$. Since $d/e = f\cdot n$ with n odd, the range $(0, fn/2)$ consists of $(n-1)/2$ full periods and one copy of $(0, f/2)$, so each inner sum is $\chi^*(e)h(\chi^*)$. This gives $H = h(\chi^*)\prod_{\ell\mid m'}(1-\chi^*(\ell))$.
- In T, the inner sums are $\bar\chi^*(e)\sum_{u \bmod fn}\bar\chi^*(u)\tan(\pi u/(fn))$ with $n = d/(ef)$ odd. Writing $u = a + fb$ and using $\sum_{b<n}\tan(y+\pi b/n) = n\tan ny$ (proof of Lemma 9.2(a)), each is $\bar\chi^*(e)\,n\,t(\chi^*)$. This gives $T = \frac{d}{f}t(\chi^*)\prod_{\ell\mid m'}(1-\bar\chi^*(\ell)/\ell)$. □

Lemma 4.1 is classical (compare B. C. Berndt, *Classical theorems on quadratic residues*, Enseign. Math. 22 (1976)). For the appearance of $|L(1,\chi)|^4$, compare Walum's formula for $\sum_h s(h,p)^2$ (H. Walum, Illinois J. Math. 26 (1982)). We know of no earlier character expansion of a limit law for $S(p)$.

Proof of Theorem 2. (a) Write $\widehat{G_d}(\chi^*) = \mathbb{E}[G_d\,\bar\chi^*]$. The continuous characters of the compact abelian group Ω are the functions $x\mapsto\chi(x \bmod f)$, for χ primitive of odd conductor f (including the trivial character, $f=1$). They form an orthonormal basis of $L^2(\Omega)$. Fix χ^* of conductor f; the coefficient of χ^* in G_d vanishes unless $f\mid d$, and is then given by Lemma 4.2. It vanishes if χ^* is even (in particular if $f=1$). Since $|h(\chi^*)|\le f$, $|t(\chi^*)|\le\frac{2f}{\pi}\cdot 3|L(1,\bar\chi^*)|$ and $d/\varphi(d)\ll\log\log d$, these coefficients are $\ll_{\chi^*} 4^{\omega(d)}\log\log d$, so $\sum_d|\widehat{G_d}(\chi^*)|/d^2<\infty$. As $R_K\to R_\infty$ in L^2 and $w_d(K)\uparrow\pi^2/8$, dominated convergence gives

$$\widehat{R_\infty}(\chi^*) = \frac{\pi^2}{8}\sum_{\substack{d\ \text{odd}\\ f\mid d}}\frac{\widehat{G_d}(\chi^*)}{d^2}.$$

Write $d = fm$ with m odd. Then $\varphi(d) = \varphi(f)\,m\prod_{\ell\mid m'}(1-1/\ell)$, and Lemma 4.2 turns the sum into $\frac{h(\chi^*)t(\chi^*)}{f^2\varphi(f)}\sum_{m\ \text{odd}} m^{-2}g(m)$. Here g is multiplicative, $g(\ell^v)=1$ for $\ell\mid f$, and $g(\ell^v) = \frac{(1-\chi^*(\ell))(1-\bar\chi^*(\ell)/\ell)}{1-1/\ell}$ for $\ell\nmid f$. The Euler product of $\sum_m g(m)m^{-2}$ is $E(\chi^*)$. Finally, Lemma 4.1 gives $h(\chi^*)t(\chi^*) = -\frac{2f\tau(\chi^*)\chi^*(2)}{\pi^2 i}(2-\bar\chi^*(2))^2L(1,\bar\chi^*)^2$, using $2\chi(2)-1=\chi(2)(2-\bar\chi(2))$. So $\widehat{R_\infty}(\chi^*) = W(\chi^*)$.

(b) Parseval's identity, with $|\tau(\chi)|^2 = f$ and $|2-\bar\chi(2)| = |2-\chi(2)|$.

(c) By Theorem 1(c), $c_0(q;r)-c_0$ is $\frac{4}{\pi}$ times the conditional mean of R_∞ given $x\equiv r \pmod{q'}$. That conditional expectation is the orthogonal projection onto the span of the characters that factor through $(\mathbb{Z}/q'\mathbb{Z})^\times$, i.e. those with $f_\chi\mid q'$. For $q=3$: $\tau(\chi_{-3}) = i\sqrt{3}$, $\chi_{-3}(2)=-1$, $L(1,\chi_{-3}) = \pi/(3\sqrt{3})$, and $E(\chi_{-3}) = \frac{9}{8}\prod_{\ell\equiv 5\,(6)}(1+2/(\ell-1)^2) = 1.318174\ldots$. □

Remark 4.3 (Numerical checks).
- Lemmas 4.1 and 4.2 agree with direct evaluation of h, t and the Fourier coefficients of G_d to 10^{-12} (all odd characters modulo $d = 9$, 15, 21, 25, 27, 35, 45, 63).
- Formula (c) reproduces every entry of Table 2 of [1] for $q = 3,5,7,9$ to the four digits printed, except $c_0(9;8) = -0.28905$, which Table 2 of [1] rounds to -0.2891 (within its stated uncertainty). For instance, it gives $c_0(5;1) = 0.5863$, $c_0(5;2) = 0.4159$, $c_0(7;1) = 0.4656$ and $c_0(9;1) = 0.7689$. [1] obtained these by summing over $d\le 8001$ and extrapolating.
- Beyond Table 2, formula (c) was tested against the primes themselves. For every odd $q\le 45$ and every r (422 classes, with predicted values from -0.621 to 1.101), the mean of X_p over the primes $10^3 < p\le 5.5\cdot 10^6$ in the class agrees with $c_0(q;r)$ to within 1.4 standard errors, and to within 0.0072 in absolute terms (`data/class_means_theoremJ_q45.txt`).
- The series in (b), summed over the conductors $f\le F$, gives 0.389678, 0.391018, 0.391467, 0.391692, 0.391827, 0.392008 and 0.392098 for $F = 401, 801, 1201, 1601, 2001, 3001, 4001$.

Its tail is $\approx a/F$, and extrapolation gives $\sigma^2 = 0.392369 \pm 0.000002$ ($\sigma = 0.626393$; the spread is over fits of the tail by a/F, $a/F + b/F^2$ and $(a + b\log F)/F$), in agreement with the value 0.3924 ± 0.0001 found above by a different method.

The code is `code/char_expansion.py`.

Corollary 4.4 (Classes with smooth moduli)**.** *Let $Q_K = \prod_{3\le\ell<K} \ell^{a_\ell}$, with a_ℓ minimal such that $\ell^{a_\ell} > K^3$. As $K \to \infty$,*

$$c_0(Q_K; \pm 1) = c_0 \pm \Big(\kappa_H \log K \,\big(\log K - \gamma - \log 2 + K_0\big) + C_Q + o(1)\Big), \qquad \kappa_H = \frac{2e^\gamma}{\pi^2},$$

where $C_Q = -\frac{4}{\pi}\sum_{e \text{ odd}} \frac{1}{e}\Big(\frac{\log e + K_0}{\pi} - \frac{F_e(1)}{e}\Big) = -0.35153\ldots$ (the term $e = 1$ has $F_1(1) = 0$).

Proof. By (9.2) and the proof of Theorem 1(b), $c_0(Q; \pm 1) - c_0 = \pm\frac{\pi}{2}\sum_{d\mid Q,\, d\ge 3} G_d(1)/d^2 + o(1)$ for $Q = Q_K$. Möbius inversion of Lemma 9.1 gives $G_d = \sum_{e\mid d} \mu(d/e) F_e$ (with $F_1 = 0$), hence

$$\sum_{d\mid Q} \frac{G_d(1)}{d^2} = \sum_{e\mid Q} \frac{F_e(1)}{e^2} \prod_{\ell \mid Q/e} \Big(1 - \frac{1}{\ell^2}\Big).$$

The product equals $\prod_{3\le\ell<K}(1 - \ell^{-2}) = \frac{8}{\pi^2}(1 + O(K^{-1}))$, unless $\ell^{a_\ell} \mid e$ for some ℓ. The e of that exceptional kind contribute $O((\log K)^2 K^{-2})$, since $\ell^{-a_\ell} < K^{-3}$. By Corollary 4.3(a), $F_e(1)/e^2 = (\log e + K_0)/(\pi e) + O(e^{-3})$. Truncating at the exponents a_ℓ changes the Euler products below by factors $1 + O(K^{-2})$. So

$$\sum_{e\mid Q} \frac{\log e + K_0}{e} = E_K\Big(\sum_{3\le\ell<K} \frac{\log \ell}{\ell - 1} + K_0\Big)(1 + O(K^{-2})), \qquad E_K = \prod_{3\le\ell<K} \Big(1 - \frac{1}{\ell}\Big)^{-1}.$$

By the prime number theorem with error term, $E_K = \frac{e^\gamma}{2}\log K\,(1 + O(e^{-c\sqrt{\log K}}))$ and $\sum_{3\le\ell<K} \frac{\log \ell}{\ell - 1} = \log K - \gamma - \log 2 + O(e^{-c\sqrt{\log K}})$. Collecting, $\frac{\pi}{2}\sum_{d\mid Q} G_d(1)/d^2 = \frac{\pi}{2}\cdot\frac{8}{\pi^2}\cdot\frac{e^\gamma}{2\pi}\log K(\log K - \gamma - \log 2 + K_0) + O(1)$, up to the $O(1)$ terms. These are the term $e = 1$ and the remainders of Corollary 4.3(a), which together sum to C_Q; every other error is $o(1)$. □

Remark 4.5. Exact values of $\frac{\pi}{2}\sum_{d\mid Q_K} G_d(1)/d^2$ are 15.89, 28.84 and 45.64 for $K = 10^3, 10^4, 10^5$. The two-term formula gives 16.10, 29.12 and 45.96: the differences $-0.21, -0.28, -0.32$ approach $C_Q = -0.3515$; the remaining drift comes from the Mertens-product error at these K. The leading term $\kappa_H(\log K)^2$ alone is off by 1.3, 1.8 and 2.2. Note $-\gamma - \log 2 + K_0 = -0.4515\ldots$.

5. Extreme values

Here we prove the extreme-value theorem, with the same constant as the tails below. The proof transfers the deterministic contribution of the small moduli, computed in Lemma 5.3, to the primes of one residue class, using Gallagher's prime number theorem. For that, only an upper bound for a mean square and a lower bound for the size of the class are needed.

Theorem 3. *For every $\varepsilon > 0$ there is $X_0(\varepsilon)$ such that for $X \ge X_0(\varepsilon)$ there are primes $p, p' \in (X, 2X]$ with*

$$X_p \le c_0 - (\kappa_H - \varepsilon)(\log\log p)^2, \qquad X_{p'} \ge c_0 + (\kappa_H - \varepsilon)(\log\log p')^2, \qquad \kappa_H = \frac{2e^\gamma}{\pi^2}.$$

In fact each inequality holds for at least $X^{1-o(1)}$ primes in $(X, 2X]$. The constant $X_0(\varepsilon)$ is effectively computable.

Thus $X_p - c_0 = \Omega_\pm((\log\log p)^2)$ unconditionally and effectively. Together with [1, Theorem E], which gives $X_p - c_0 = O((\log\log p)^2)$ for all p outside a set of density zero, this shows that $(\log\log p)^2$ is the true order of the extreme deviations.

5.1. **External input.** We use the following two results, for Dirichlet characters to moduli $r \le R$. A real zero $\tilde\beta$ of $L(s,\tilde\chi)$, with $\tilde\chi$ primitive of conductor $\tilde q \le R$, is *exceptional of level R and quality δ* if $\tilde\beta \ge 1 - \delta/\log R$.

(LP) (Landau–Page.) There is an absolute $\delta_0 \in (0,1)$ such that for all large R there is at most one exceptional zero of level R and quality δ_0. If it exists, $\tilde\chi$ is real and $\tilde\beta$ is simple.

(Ga) (Gallagher, *Invent. Math.* 11 (1970), 329–339, Theorem 7.) There are absolute constants $b_G, c > 0$ such that the following holds for $\exp((\log N)^{1/2}) \le R \le N^{b_G}$ and $0 < \delta < 1$:

$$\sum_{r\le R}\sideset{}{^*}\sum_{\chi \bmod r}\max_{I\subset[1,N]}\frac{1}{|I|+N/R}\Big|\sum_{n\in I}\big(\Lambda(n)\chi(n)-\mathbf{1}_{r=1}+\mathbf{1}_{\chi=\tilde\chi}\,n^{\tilde\beta-1}\big)\Big| \ll E_\delta(N,R).$$

Here $E_\delta = \exp(-c\delta\log N/\log R)$ if there is no exceptional zero of level R and quality δ, and then the term with $\tilde\chi$ is omitted. Otherwise $E_\delta = \exp(-c\log N/\log R)$. (We use the formulation of Lemma 7.3 of L. Grimmelt and J. Teräväinen, arXiv:2508.16400, but only in Gallagher's range $R \le N^{b_G}$. The range $R \le N$ stated there is too generous near $R = N$; their paper uses only $R \le N^{1/7}$.)

With $\delta = \delta_0$, both cases give $E_{\delta_0} \ll \exp(-c_2\log N/\log R)$ with $c_2 = c\delta_0$.

5.2. **Primes in a class, tested against periodic functions.** For odd Q and $(a,Q)=1$ write $\mathbb{E}[\cdot \mid a] = \mathbb{E}[\cdot \mid x \equiv a \pmod Q]$ on Ω. Let $\mathcal{P} = \mathcal{P}_X(a,Q)$ be the set of primes $p \in (X, 2X]$ with $p \equiv a \pmod Q$.

Lemma 5.1. *Let X be large and $\exp(\sqrt{\log 2X}) \le T \le X^{b_G}$, where we may assume $b_G \le \frac14$. Let Q be odd with $(a,Q)=1$, and let f be a function on the reduced residues modulo an odd M with $\operatorname{lcm}(M,Q) \le T$. Then*

$$\sum_{p\in\mathcal{P}} f(p)\log p = \frac{X}{\varphi(Q)}\Big(\mathbb{E}[f \mid a] - \theta\,\mathbb{E}[f\tilde\chi \mid a]\Big) + O\Big(\frac{\|f\|_\infty X}{\varphi(Q)}e^{-c_2\log X/\log T}\Big).$$

If there is an exceptional zero $\tilde\beta$ of level T and quality δ_0, then $\tilde\chi$ is its character and $\theta = X^{-1}\sum_{X<n\le 2X} n^{\tilde\beta-1} \in (0,1]$; otherwise $\theta = 0$. The number θ depends only on X and T.

Proof. Put $M' = \operatorname{lcm}(M,Q)$ and $I = (X,2X]$. Expanding $f\cdot\mathbf{1}_{x\equiv a(Q)}$ in characters modulo M',

$$\sum_{\substack{n\in I\\ n\equiv a(Q)}}\Lambda(n)f(n) = \sum_{\chi \bmod M'}\hat f(\chi)\sum_{n\in I}\Lambda(n)\chi(n), \qquad \hat f(\chi) = \frac{1}{\varphi(M')}\sum_{\substack{x \bmod M'\\ x\equiv a(Q)}} f(x)\bar\chi(x).$$

The inner sum has $\varphi(M')/\varphi(Q)$ terms, so $|\hat f(\chi)| \le \|f\|_\infty/\varphi(Q)$.

Each $\chi \bmod M'$ is induced by a unique primitive χ^* of conductor $r \mid M'$, $r \le T$, and the two sums over $n \in I$ differ by $O(\log M' \log X)$. By (Ga) with $N = 2X$, $R = T$ (so $|I| + N/R \le 3X$), and by $\sum_{n\in I} n^{\tilde\beta-1} = \theta X$:

- the principal character contributes $\hat f(\chi_0)X = \frac{X}{\varphi(Q)}\mathbb{E}[f \mid a]$;
- the character induced by $\tilde\chi$, present only when $\tilde q \mid M'$, contributes $-\hat f(\tilde\chi)\theta X = -\frac{\theta X}{\varphi(Q)}\mathbb{E}[f\tilde\chi \mid a]$;
- all characters together contribute an error $O(\|f\|_\infty\varphi(Q)^{-1}(Xe^{-c_2\log X/\log T} + T\log^2 X))$.

If $\tilde q \nmid M'$, we set $\mathbb{E}[f\tilde\chi \mid a] = 0$. This is consistent: when $\tilde q$ is odd, the expectation computed modulo $\operatorname{lcm}(M',\tilde q)$ vanishes; indeed, $\tilde q$ is then squarefree, so some prime $\ell \mid \tilde q$ does not divide M'. (When $4 \mid \tilde q$ the character never occurs in the expansion modulo the odd number M'.) The corresponding local component of x is uniform and independent of $x \bmod M'$ on $\{x \equiv a\ (Q)\}$, and the local component of $\tilde\chi$ is nontrivial. So the formula holds in all cases.

Prime powers contribute $O(\|f\|_\infty X^{1/2}\log X)$, and $T\log^2 X + X^{1/2}\log X \ll Xe^{-c_2\log X/\log T}/\varphi(Q)$ since $\varphi(Q)\le T\le X^{1/4}$ and $e^{-c_2\log X/\log T}\ge e^{-c_2\sqrt{\log X}}$. □

5.3. **The random part is small in mean square.** Fix $K\ge 5$, $Q=Q_K=\prod_{3\le\ell<K}\ell^{a_\ell}$ (with a_ℓ minimal such that $\ell^{a_\ell}>K^3$), any a coprime to Q, and $Y>Q$. Put

$$R'_a(x)=\sum_{\substack{3\le d<Y\\ d\nmid Q}}\frac{w_d(Y)}{d^2}G_d(x).$$

Lemma 5.2. *$\mathbb{E}[R'^2_a\mid a]\le\eta(K)$, where $\eta(K)\to 0$ as $K\to\infty$ uniformly in Y and a.*

Proof. Under $\mathbb{P}(\,\cdot\mid a)$ the components of x at the various primes are independent. At $\ell<K$ the residue x mod ℓ^{a_ℓ} is fixed and the higher ℓ-adic digits are uniform; at $\ell\ge K$ the unit $x\in\mathbb{Z}_\ell^\times$ is Haar distributed.

For d let $\mathcal{S}(d)$ be the set of primes $\ell\mid d$ at which $x \bmod \ell^{v_\ell(d)}$ is not determined by the condition. These are the $\ell\ge K$, and the $\ell<K$ with $v_\ell(d)>a_\ell$. Then $d\nmid Q$ if and only if $\mathcal{S}(d)\ne\emptyset$. For a pair d,d' put $\mathcal{S}_0=\mathcal{S}(d)\cap\mathcal{S}(d')$, and let ξ be the $\mathcal{S}_0$-components of x.

Conditionally on ξ, the values $G_d(x)$ and $G_{d'}(x)$ depend on disjoint sets of independent components. So

$$\mathbb{E}[G_dG_{d'}\mid a]=\mathbb{E}_\xi\big[\mathbb{E}[G_d\mid a,\xi]\,\mathbb{E}[G_{d'}\mid a,\xi]\big],$$

and this equals $\mathbb{E}[G_d\mid a]\mathbb{E}[G_{d'}\mid a]$ if $\mathcal{S}_0=\emptyset$. Moreover $\mathbb{E}[G_d\mid a,\xi]=A^*_{d,g}(s)$ for some s, where $g=g_d\mid d$ is the part of d at which x is determined by a and ξ; explicitly $g_d=(d_0,Q)\,d_1$ in the notation below. By Lemma 9.2(c) (and $A^*_{d,1}=0$) this has absolute value at most $C\tau(d)^2\lambda(d)\,g_d$.

Write $c_d=w_d(Y)/d^2\le\frac{\pi^2}{8}d^{-2}$. Then

$$\mathbb{E}[R'^2_a\mid a]\le\Big(\sum_{d\nmid Q}|c_d|\,|\mathbb{E}[G_d\mid a]|\Big)^2+C^2\sum_{\mathcal{S}_0\ne\emptyset}|c_dc_{d'}|\,\tau(d)^2\tau(d')^2\lambda(d)\lambda(d')\,g_dg_{d'}.$$

The first term. $\mathbb{E}[G_d\mid a]=A^*_{d,(d,Q)}(a)$, so the first term tends to 0 by the three-case estimate in the proof of Theorem 1(b). That estimate uses only Lemma 9.2(c), so it is uniform in a.

The second term. For a pair with $\mathcal{S}_0\ne\emptyset$, write $d=d_0d_1$, where $d_1=\prod_{\ell\in\mathcal{S}_0}\ell^{v_\ell(d)}$ is the full $\mathcal{S}_0$-part of d. Then $g_d\le(d_0,Q)\,d_1$. Use $\tau(mn)\le\tau(m)\tau(n)$, $\lambda(mn)\le\lambda'(m)\lambda'(n)$ and $\lambda'(mn)\le 2\lambda'(m)\lambda'(n)$, and write $d_0=(d_0,Q)\,u$. Then the contribution of d is at most

$$\frac{\tau(d)^2\lambda(d)\,g_d}{d^2}\le\frac{\tau((d_0,Q))^2\lambda'((d_0,Q))}{(d_0,Q)}\cdot\frac{\tau(u)^2\lambda'(u)}{u^2}\cdot\frac{\tau(d_1)^2\lambda'(d_1)}{d_1}.$$

For $\ell\in\mathcal{S}_0$ we have $v_\ell(d)\ge 1$ if $\ell\ge K$, and $v_\ell(d)\ge a_\ell+1$ if $\ell<K$. Summing over all pairs, the second term is at most

$$C\Big(\sum_{g\mid Q}\frac{\tau(g)^2\lambda'(g)}{g}\Big)^2\Big(\sum_{u\ge1}\frac{\tau(u)^2\lambda'(u)}{u^2}\Big)^2\sum_{\mathcal{S}_0\ne\emptyset}\prod_{\ell\in\mathcal{S}_0}\Big(2\sum_{v\ge m_\ell}\frac{(v+1)^2\lambda'(\ell^v)}{\ell^v}\Big)^2,$$

with $m_\ell=1$ for $\ell\ge K$ and $m_\ell=a_\ell+1$ for $\ell<K$.

- The first factor is $(\log K)^{O(1)}$ (as in the proof of Theorem 1(b), first bullet, using $\lambda'(g)\ll(\log K)^3g^{1/\log K}$).
- The second factor is $O(1)$.
- Each local factor is $\ll(\log\ell)^3/\ell^2$ for $\ell\ge K$, and $\ll(\log K)^{O(1)}K^{-6}$ for $\ell<K$ (since $\ell^{a_\ell}>K^3$). So the last sum is at most $\exp\big(C\sum_{\ell\ge K}(\log\ell)^3\ell^{-2}+(\log K)^{O(1)}K^{-5}\big)-1\ll(\log K)^3K^{-1}$.

Hence the second term is $\ll(\log K)^{O(1)}K^{-1}\to 0$. □

5.4. **The deterministic part.**

Lemma 5.3. *Let $\epsilon \in \{1,-1\}$, $Q = Q_K$, and $a \equiv \epsilon \pmod Q$. Put $D(a) = \sum_{d\mid Q,\, d\ge 3} w_d(Y) G_d(a)/d^2$ with $Y > Q$. Then $\epsilon D(a) \ge (\frac{e^\gamma}{2\pi} - o(1))(\log K)^2$ as $K \to \infty$.*

Proof. Since $a \equiv \epsilon \pmod d$ for all $d \mid Q$, we have $\epsilon D(a) = \sum_{d\mid Q} w_d(Y) G_d(1)/d^2$, and every $G_d(1)$ is positive. For $m \mid Q/d$ we have $dm \le Q < Y$, so $w_d(Y) \ge \sum_{m\mid Q/d} m^{-2}$. By Lemma 9.1 (the divisors of Q are closed under taking divisors) and Corollary 4.3(a), this is at least

$$\sum_{k\mid Q,\, k\ge 3} \frac{F_k(1)}{k^2} \ge \frac{1}{\pi} \sum_{k\mid Q} \frac{\log k}{k} - 1.$$

Next,

$$\sum_{k\mid Q} \frac{\log k}{k} \;\ge\; \sum_{3\le \ell < K} \log \ell \sum_{\substack{k\mid Q \\ \ell \mid k}} \frac{1}{k} \;\ge\; \sum_{3\le \ell < K} \frac{\log \ell}{\ell}\Big(1 - \frac{1}{\ell}\Big) \prod_{3\le \ell' < K} \sum_{b \le a_{\ell'}} \ell'^{-b}.$$

The product is at least $(1-K^{-2}) \prod_{3\le \ell' < K} (1 - 1/\ell')^{-1} = (\frac{e^\gamma}{2} + o(1)) \log K$ by Mertens. So $\epsilon D(a) \ge (\frac{e^\gamma}{2\pi} + o(1))(\log K)^2$. □

5.5. **Proof of Theorem 3.** Let c_2 be as after (Ga), let X be large, and put

$$\log T_0 = \frac{c_2 \log X}{20 \log\log X}.$$

Let K be maximal such that $Q := Q_K$ satisfies $Q^3 (\log X)^{32} \le T_0$, and put

$$Y = Q(\log X)^{16}, \qquad T = QY^2 \le T_0.$$

Since $\log Q_K = (4 + o(1))K$, we get $K = (\frac{c_2}{240} + o(1)) \log X / \log\log X$, hence $\log K = (1 + o(1)) \log\log X$. Also $T \ge \exp(\sqrt{\log X})$ and $Q = X^{o(1)}$.

Choice of the class. Fix $\epsilon = -1$ for p, or $\epsilon = +1$ for p'.

- If there is no exceptional zero of level T and quality δ_0, or if its conductor $\tilde q$ does not divide Q, take $a \equiv \epsilon \pmod Q$.
- Otherwise $\tilde q \mid Q$. This exceptional case is treated in Section 5.6.

In the first case, Lemma 5.3 gives

$$\epsilon D(a) \ge \Big(\frac{e^\gamma}{2\pi} - o(1)\Big)(\log K)^2.$$

Decomposition. Let $p \in \mathcal{P} = \mathcal{P}_X(a, Q)$ and $K_2 = \sqrt{2X}(\log 2X)^4$. By Theorem B(ii) and Lemma 9.1,

$$X_p - c_0 = \frac{4}{\pi}\Big(D(a) + R'_a(p) + V(p)\Big) + O\Big(\frac{1}{\log X}\Big), \qquad V = R_{K_2} - R_Y,$$

because $G_d(p) = G_d(a)$ for $d \mid Q$.

Size of the class. By Lemma 5.1 with $f \equiv 1$ and $M = 1$,

$$\sum_{p\in\mathcal{P}} \log p = \frac{X}{\varphi(Q)} + O\Big(\frac{X}{\varphi(Q)} (\log X)^{-20}\Big) \ge (1 - o(1)) \frac{X}{\varphi(Q)},$$

since $c_2 \log X / \log T \ge 20 \log\log X$ (as $\tilde q \nmid Q$, there is no exceptional term).

Second moment of R'_a. Apply Lemma 5.1 to each $f = G_d G_{d'}$ ($d, d' < Y$, $d, d' \nmid Q$, $M = \operatorname{lcm}(d, d')$, $\operatorname{lcm}(M, Q) \le T$), multiply by $c_d c_{d'}$ and sum. The main terms add up to $\frac{X}{\varphi(Q)} \mathbb{E}[R'^2_a (1 -$

$\theta\tilde{\chi}) \mid a] \le \frac{2X}{\varphi(Q)}\mathbb{E}[R_a'^2 \mid a]$, because $R_a'^2 \ge 0$ and $0 \le \theta \le 1$. By Lemma 9.2(b), $\sum_{d<Y} |c_d| \, \|G_d\|_\infty \ll (\log Y)^3$, so the errors add up to $O(X\varphi(Q)^{-1}(\log Y)^6(\log X)^{-20})$. With Lemma 5.2:

$$\sum_{p\in\mathcal{P}} R_a'(p)^2 \log p \le \frac{2X}{\varphi(Q)}\eta(K) + o\Big(\frac{X}{\varphi(Q)}\Big) = o\Big(\frac{X}{\varphi(Q)}\Big).$$

The tail V. By Lemmas 9.3(b) and 9.4 over all integers,

$$\sum_{p\in\mathcal{P}} V(p)^2 \log p \le \log(2X) \sum_{X<n\le 2X} V(n)^2 \ll \frac{X(\log X)^{15}}{Y} = \frac{X}{Q\log X} = o\Big(\frac{X}{\varphi(Q)}\Big).$$

Conclusion. The log-weighted proportion of $p \in \mathcal{P}$ with $|R_a'(p) + V(p)| > 1$ is $o(1)$. For all other $p \in \mathcal{P}$,

$$\epsilon(X_p - c_0) \ge \frac{4}{\pi}\Big(\frac{e^\gamma}{2\pi} - o(1)\Big)(\log K)^2 - O(1) = (\kappa_H - o(1))(\log\log p)^2.$$

Such primes number at least $(1-o(1))X/(\varphi(Q)\log X) = X^{1-o(1)}$. This proves Theorem 3 unless $\tilde{q} \mid Q$; that case is treated in Section 5.6.

Remark 5.4. *Context.* The method is a transfer of standard tools: Gallagher's prime number theorem at moduli $\exp(c\log X/\log\log X)$, a mean-square bound so that only an upper bound is needed, and, for the exceptional character, the removal of one prime from Q (Section 5.6), which replaces Siegel's theorem. An analogous unconditional result for $L(1,\chi)$ is due to Joshi (P. T. Joshi, *J. Number Theory* 2 (1970), 58–73): $L(1,(\cdot/q)) \ge (e^\gamma - \varepsilon)\log\log q$ for infinitely many primes q.

Why GRH is not needed. An asymptotic for each class separately would need GRH. Here only an *upper* bound for a nonnegative quantity is needed, together with a lower bound for the size of one class. Gallagher's theorem gives both with relative error $\exp(-c\log X/\log T)$, which is a negative power of $\log X$ once $\log T \asymp \log X/\log\log X$. This modulus is still large enough that $\log K \sim \log\log X$.

5.6. **The exceptional case.** It remains to treat the case where there is an exceptional zero $\tilde{\beta}$ of level T and quality δ_0 whose conductor $\tilde{q}$ divides Q. The usual tool here is Siegel's theorem, which is ineffective. Instead we remove one prime from Q.

Proof of Theorem 3 in the exceptional case. Let ℓ_1 be the largest prime factor of $\tilde{q}$, which is odd and squarefree.

ℓ_1 is large, effectively. For a real primitive character of conductor $\tilde{q}$, every real zero satisfies $1 - \tilde{\beta} \ge c_P\,\tilde{q}^{-1/2}(\log\tilde{q})^{-2}$, with $c_P > 0$ effective [3, Chapter 14]. With $1 - \tilde{\beta} \le \delta_0/\log T$ this gives $\log\tilde{q} \ge (2-o(1))\log\log T$. Also $\tilde{q} \le \prod_{\ell\le\ell_1}\ell \le e^{1.02\ell_1}$ and $\log\log T = (1+o(1))\log K$, so $\ell_1 \ge (1.9+o(1))\log K$; in particular $\ell_1 \to \infty$ (only this is needed below).

The class. Put $Q' = Q/\ell_1^{a_{\ell_1}}$ and take $a \equiv \epsilon \pmod{Q'}$. Since $\ell_1 \nmid Q'$, $\tilde{q} \nmid Q'$. Let $\mathcal{P}' = \mathcal{P}_X(a, Q')$. By Lemma 5.1 with $f \equiv 1$ and $M = 1$, there is no exceptional term, and $\sum_{p\in\mathcal{P}'}\log p = (1+o(1))X/\varphi(Q')$.

Decomposition. With $c_d = w_d(Y)/d^2$, for $p \in \mathcal{P}'$,

$$R_Y(p) = D'(a) + U(p) + R''(p), \qquad D'(a) = \sum_{d\mid Q',\, d\ge 3} c_d G_d(a), \quad U = \sum_{\substack{d\mid Q\\ \ell_1\mid d}} c_d G_d, \quad R'' = \sum_{\substack{3\le d<Y\\ d\nmid Q}} c_d G_d.$$

- D'. Every $d \mid Q'$ satisfies $a \equiv \epsilon \pmod d$, so $G_d(a) = \epsilon G_d(1)$. The argument for the first sum in the proof of Lemma 5.3, applied to Q' in place of Q, gives $\epsilon D'(a) \ge \frac{1}{\pi}\sum_{k\mid Q'}\frac{\log k}{k} - 1$. Compared with Q, the Mertens product loses the factor $1 - 1/\ell_1$,

and the sum over ℓ loses the term $\ell = \ell_1$, which is at most $\frac{\log \ell_1}{\ell_1}$. Since $\ell_1 \to \infty$, $\epsilon D'(a) \geq (\frac{e^\gamma}{2\pi} - o(1))(\log K)^2$.

- *U, pointwise.* By Lemma 6.3 below, $G_d(1) \leq F_d(1)$ (Lemma 9.1, positivity) and Corollary 4.3(b),

$$|U| \leq \frac{\pi^2}{8} \sum_{\substack{d|Q \\ \ell_1|d}} \frac{\log d + K_0}{\pi d} \ll \frac{1}{\ell_1} \sum_{g|Q'} \frac{\log g + \log \ell_1 + 1}{g} \ll \frac{(\log K)^2}{\ell_1} = o\big((\log K)^2\big).$$

- *R'', in mean square.* Apply Lemma 5.1 with Q' in place of Q to $f = G_d G_{d'}$ ($d, d' < Y$, $d, d' \nmid Q$). As in Section 5.5, the main terms add up to $\frac{X}{\varphi(Q')}\mathbb{E}[R''^2(1 - \theta\tilde{\chi}) \mid a] \leq \frac{2X}{\varphi(Q')}\mathbb{E}[R''^2 \mid a]$, where now the condition is $x \equiv a \pmod{Q'}$. Conditioning further on $x \bmod \ell_1^{a_{\ell_1}}$ fixes $x \bmod Q$, and R'' is the function $R'_{a'}$ of Lemma 5.2 for the resulting class $a' \bmod Q$. Since that lemma is uniform in a', $\mathbb{E}[R''^2 \mid a] \leq \eta(K)$. The error terms are as before.
- *The tail $V = R_{K_2} - R_Y$* is unchanged.

As in Section 5.5, all but a log-weighted proportion $o(1)$ of $p \in \mathcal{P}'$ satisfy $|R''(p) + V(p)| \leq 1$. For them,

$$\epsilon(X_p - c_0) \geq \frac{4}{\pi}\Big(\epsilon D'(a) - |U(p)| - 1\Big) + O\Big(\frac{1}{\log X}\Big) \geq (\kappa_H - o(1))(\log\log p)^2,$$

and there are $X^{1-o(1)}$ of them. Every $o(1)$ here is effective. □

Remark 5.5 (Effectivity). All the inputs are effective: (LP); Gallagher's theorem (Ga), whose constants b_G, c are effective because his proof uses only the log-free zero-density estimate and the Deuring–Heilbronn phenomenon; the prime number theorem (for $\log Q_K$) and Mertens' theorems; Theorem B(ii); and Lemmas 9.2–9.4, 5.2 and 5.3, whose constants are explicit. Hence the constant $X_0(\varepsilon)$ in Theorem 3 is effectively computable.

6. Tails, moments, and a conjecture

6.1. **Tails of the limiting distribution.** Lemmas 5.2 and 5.3 also describe the tails of the law ν of Theorem 1.

Corollary 6.1. *As $t \to \infty$,*

$$\nu\big((-\infty, c_0 - t)\big) \;\geq\; \exp\Big(-\exp\big((1 + o(1))\sqrt{t/\kappa_H}\big)\Big),$$

and the same lower bound holds for $\nu\big((c_0 + t, \infty)\big)$. Consequently, as $t \to \infty$, the primes with $S(p) < \frac{p}{2}\log p + (c_0 - t)p$, and also the primes with $S(p) > \frac{p}{2}\log p + (c_0 + t)p$, have lower density at least $\exp(-\exp((1 + o(1))\sqrt{t/\kappa_H}))$.

Proof. Let $\epsilon = \mp 1$, and let $\mathcal{C} = \{x \equiv \epsilon \pmod{Q_K}\}$, so that $\mathbb{P}(\mathcal{C}) = 1/\varphi(Q_K) \geq e^{-(4+o(1))K}$. On $\mathcal{C}$ we have

$$R_\infty = D_\infty + R'_\infty, \qquad D_\infty = \frac{\pi^2}{8} \sum_{d|Q_K,\, d \geq 3} \frac{G_d(\epsilon)}{d^2}.$$

Here R'_∞ is the L^2 limit, as $Y \to \infty$, of R'_a in Lemma 5.2 with $a = \epsilon$; it exists by Lemma 3.1.

- By Lemma 5.2, $\mathbb{E}[R'^2_\infty \mid \mathcal{C}] \leq \eta(K) \to 0$, so $\mathbb{P}(|R'_\infty| \leq 1 \mid \mathcal{C}) \geq \frac{1}{2}$ for large K.
- By Lemma 5.3, and $w_d \to \pi^2/8$, we have $\epsilon D_\infty \geq (\frac{e^\gamma}{2\pi} - o(1))(\log K)^2$.

Hence $\epsilon \cdot \frac{4}{\pi} R_\infty \geq (\kappa_H - o(1))(\log K)^2 - \frac{4}{\pi}$ on a subset of $\mathcal{C}$ of measure at least $\frac{1}{2} e^{-(4+o(1))K}$. Given t, for large t let K be the least integer ≥ 5 with $(\kappa_H - f(K))(\log K)^2 - \frac{4}{\pi} > t$, where $f(K) \to 0$ is the $o(1)$ above, so that $\log K = (1+o(1))\sqrt{t/\kappa_H}$. The statement about primes follows from Theorem 1(a) by the portmanteau theorem. □

Remark 6.2. This is the analogue of the double-exponential tails known for the distribution of $L(1,\chi)$ (A. Granville and K. Soundararajan, *Geom. Funct. Anal.* 13 (2003), 992–1028). There the tails are $\exp(-\exp(Ct))$; here the exponent is $\sqrt{t}$, because the extreme values of X_p are of order $(\log\log p)^2$ rather than $\log\log p$. Theorem 4 below shows that the lower bound of Corollary 6.1 is sharp.

6.2. **Sharp tails.**

Theorem 4. *Let Z have the law ν of Theorem 1, and $\kappa_H = 2e^\gamma/\pi^2$.*

(a) *As $t \to \infty$,*

$$\nu\big((c_0+t,\infty)\big) = \nu\big((-\infty, c_0 - t)\big) = \exp\Big(-\exp\big((1+o(1))\sqrt{t/\kappa_H}\big)\Big).$$

(b) *$\mathbb{E}[e^{\theta|Z|}] < \infty$ for every $\theta > 0$, and as $\theta \to +\infty$,*

$$\log \mathbb{E}\big[e^{\pm\theta(Z-c_0)}\big] = (\kappa_H + o(1))\,\theta(\log\theta)^2.$$

(c) *For fixed $t > 0$ let $\underline{\delta}_\pm(t)$ and $\overline{\delta}_\pm(t)$ be the lower and upper densities, in the primes, of $\{p : \pm(X_p - c_0) > t\}$. Then, as $t \to \infty$, all four are $\exp(-\exp((1+o(1))\sqrt{t/\kappa_H}))$.*

The lower bounds in (a) are Corollary 6.1. The upper bounds come from Lemma 6.5 below, which separates the small primes, whose contribution is bounded deterministically, from the large primes, whose contribution is a martingale with small increments.

Lemma 6.3. *Let $d \geq 3$ be odd and $(n,d) = 1$. Then $|G_d(n)| \leq G_d(1)$.*

Proof. Let J_d be the set of $j \in (0, d/2)$ with $(j,d) = 1$. For $j \in J_d$ let $r(j) \in J_d$ be the residue with $nj \equiv \pm r(j) \pmod d$; it exists and is unique because d is odd and $(nj,d) = 1$. The map $j \mapsto r(j)$ is injective: $r(j) = r(j')$ gives $nj \equiv \pm nj'$, so $j \equiv \pm j' \pmod d$, and $j = j'$ since both lie in $(0, d/2)$. So it is a bijection of J_d. By the π-periodicity and oddness of tan, $\tan(\pi nj/d) = \pm\tan(\pi r(j)/d)$. All the numbers $\tan(\pi r/d)$, $r \in J_d$, are positive, so $|G_d(n)| \leq \sum_{r\in J_d} \tan(\pi r/d) = G_d(1)$. □

Put $\lambda'(n) = (1 + \log\log 3n)(\log n + 2)$ as before. For $y \geq 3$ we use the majorants

$$B(y) = \sum_{g \text{ odd},\ P^+(g) < y} \frac{\tau(g)^2 \lambda'(g)}{g}, \qquad H(z) = \sum_{\substack{h>1 \\ p\mid h \Rightarrow p \geq z}} \frac{\tau(h)^2\lambda'(h)}{h^2}, \qquad H = \sum_{h\geq 1} \frac{\tau(h)^2\lambda'(h)}{h^2},$$

where $P^+(g)$ is the largest prime factor of g ($P^+(1) = 1$).

Lemma 6.4. *There is an absolute constant B_0 such that $B(y) \ll (\log y)^{B_0}$ and, for every odd prime ℓ,*

$$B_\ell := \sum_{\substack{g \text{ odd},\ P^+(g)\leq \ell \\ \ell \mid g}} \frac{\tau(g)^2\lambda'(g)}{g} \ll \frac{(\log \ell)^{B_0}}{\ell}.$$

Moreover $H < \infty$ and $H(z) \ll z^{-1}(\log z)^6$.

Proof. Let $\eta_y = 1/(4\log y)$ and let g be odd with $P^+(g) < y$. From $\log u \le u$ $(u \ge 1)$ we get $\log g \le g^{\eta_y}/\eta_y$, so $\log g + 2 \le (4\log y + 2)g^{\eta_y}$. Also $1 + \log\log 3g \le 3 + \log\log y + \eta_y \log g \le (3 + \log\log y)g^{\eta_y}$, using $a + b \le ae^b$ for $a \ge 1$, $b \ge 0$. Hence $\lambda'(g) \ll (\log y)^2 g^{2\eta_y}$. For $\ell < y$ we have $\ell^{2\eta_y} \le e^{1/2}$, so $u_\ell := \ell^{-(1-2\eta_y)} \le e^{1/2}/\ell \le 0.55$, and

$$B(y) \ll (\log y)^2 \prod_{3\le\ell<y} \sum_{b\ge0} (b+1)^2 u_\ell^b = (\log y)^2 \prod_{3\le\ell<y} \frac{1+u_\ell}{(1-u_\ell)^3} \le (\log y)^2 \exp\Big(C\sum_{\ell<y}\frac{1}{\ell}\Big) \ll (\log y)^{B_0}.$$

For B_ℓ take $y = \ell + 1$; the local factor at ℓ is now $\sum_{b\ge1}(b+1)^2 u_\ell^b \ll u_\ell \ll 1/\ell$, and the other factors are as before. Finally $H < \infty$ is clear, and every h counted in $H(z)$ is $\ge z$, so $H(z) \le \sum_{h\ge z} \tau(h)^2 \lambda'(h) h^{-2} \ll z^{-1}(\log z)^6$ by partial summation from $\sum_{h\le u}\tau(h)^2 \ll u(\log u)^3$. □

Lemma 6.5. *There are absolute constants $c, B_1 > 0$ and a function $\rho_H(z) \to 0$ such that for $z \ge 3$ and $u > 0$, with $\mathcal{Y} = \frac{4}{\pi} R_\infty$,*

$$\mathbb{P}\Big(|\mathcal{Y}| > (\kappa_H + \rho_H(z))(\log z)^2 + u\Big) \le 2\exp\Big(-c\,u^2\,\frac{z}{(\log z)^{B_1}}\Big).$$

Proof. Let $3 = \varpi_1 < \varpi_2 < \cdots$ be the odd primes and $\mathcal{F}_j$ the σ-algebra generated by the components $x_{\varpi_1}, \dots, x_{\varpi_j}$ of $x \in \Omega$ ($\mathcal{F}_0$ trivial). For odd d let $g_j(d) = \prod_{i\le j} \varpi_i^{v_{\varpi_i}(d)}$.

Conditional expectations. Conditionally on $\mathcal{F}_j$, the components of x at the primes dividing $d/g_j(d)$ are independent and Haar distributed, so $x \bmod d$ is uniform over the reduced residues modulo d lying over $x \bmod g_j(d)$. Hence $\mathbb{E}[G_d \mid \mathcal{F}_j] = A^*_{d,g_j(d)}(x)$, with $A^*_{d,d} = G_d$ and $A^*_{d,1} = 0$. By Lemma 9.2(c) and Lemma 9.2(b),

$$|A^*_{d,g}(s)| \le C\tau(d)^2\lambda(d)\,g \qquad (g \mid d), \tag{3}$$

and $\lambda(gh) \le \lambda'(g)\lambda'(h)$, $\tau(gh) = \tau(g)\tau(h)$ for coprime g, h. Writing $d = gh$ with $g = g_j(d)$, (3) and Lemma 6.4 give $\sum_d \sup_x |A^*_{d,g_j(d)}(x)|/d^2 \le C\,B(\varpi_j+1)H < \infty$. Since conditional expectation is a contraction of L^2, and $R_K \to R_\infty$ in L^2 (Lemma 3.1) with $R_K = \sum_{d<K} w_d(K) G_d/d^2$ and $w_d(K) \uparrow \pi^2/8$ (Lemma 9.1), dominated convergence gives

$$M_j := \mathbb{E}[\mathcal{Y} \mid \mathcal{F}_j] = \frac{\pi}{2} \sum_{d\ge3 \text{ odd}} \frac{A^*_{d,g_j(d)}(x)}{d^2},$$

the series converging absolutely and uniformly in x. (M_j) is an L^2-bounded martingale, and $\mathcal{Y}$ is measurable with respect to the σ-algebra generated by all the $\mathcal{F}_j$. So $M_j \to \mathcal{Y}$ in L^2.

Increments. $g_j(d) = g_{j-1}(d)$ unless $\varpi_j \mid d$. So $\Delta_j := M_j - M_{j-1}$ involves only d divisible by ϖ_j, and by (3) (with $g_{j-1}(d) \le g_j(d)$) and Lemma 6.4,

$$|\Delta_j| \le \pi C \sum_{\varpi_j \mid d} \frac{\tau(d)^2\lambda(d) g_j(d)}{d^2} \le \pi C\, B_{\varpi_j} H \le a_j := C' \frac{(\log\varpi_j)^{B_0}}{\varpi_j}.$$

The head. Let J be the number of odd primes below z, so $g_J(d)$ is the z-smooth part of d (primes $< z$). Split M_J according to whether d is z-smooth. If it is, $A^*_{d,d} = G_d$, and $|G_d(x)| \le G_d(1)$ by Lemma 6.3. If it is not, $d = gh$ with $h > 1$ composed of primes $\ge z$, and (3) bounds the total by $\frac{\pi}{2} C\,B(z)H(z) \ll z^{-1}(\log z)^{B_0+6}$. Hence, for every x,

$$|M_J| \le \frac{\pi}{2} S_z + O\big(z^{-1}(\log z)^{B_0+6}\big), \qquad S_z = \sum_{\substack{d\ge3 \text{ odd} \\ P^+(d)<z}} \frac{G_d(1)}{d^2}.$$

To evaluate S_z, sum Lemma 9.1 over the z-smooth odd $k \ge 3$ (all terms are positive): with $P_z = \prod_{3\le \ell<z}(1-\ell^{-2})^{-1}$,

$$P_z\, S_z = \sum_{\substack{k\ge 3 \text{ odd}\\ P^+(k)<z}} \frac{F_k(1)}{k^2} \le \frac{1}{\pi} \sum_{\substack{k \text{ odd}\\ P^+(k)<z}} \frac{\log k + K_0}{k} = \frac{E_z}{\pi}\Big(\sum_{3\le \ell<z} \frac{\log \ell}{\ell-1} + K_0\Big),$$

by Corollary 4.3(b) (including $k=1$ only increases the right side, since $K_0>0$), where $E_z = \prod_{3\le\ell<z}(1-1/\ell)^{-1}$ (the last step is logarithmic differentiation of $\prod(1-\ell^{-s})^{-1}$ at $s=1$). By Mertens' theorems, $E_z = (\frac{e^\gamma}{2}+o(1))\log z$ and $\sum_{\ell<z}\frac{\log\ell}{\ell-1} = \log z + O(1)$; and $P_z \to \prod_{\ell\ge 3}(1-\ell^{-2})^{-1} = \frac{\pi^2}{8}$. Hence

$$\frac{\pi}{2} S_z \le \frac{\pi}{2}\cdot\frac{8}{\pi^2}\cdot\frac{e^\gamma}{2\pi}(1+o(1))(\log z)^2 = (\kappa_H + o(1))(\log z)^2,$$

and $|M_J| \le (\kappa_H + \rho_H(z))(\log z)^2$ everywhere, for a suitable $\rho_H(z)\to 0$. (This is sharp: a closer evaluation gives $\frac{\pi}{2}S_z = \kappa_H \log z\,(\log z - \gamma - \log 2 + K_0 + o(1))$, and numerically $\frac{\pi}{2}S_z/(\kappa_H(\log z)^2) = 0.891,\ 0.943,\ 0.963,\ 0.973$ for $z = 10^2, 10^4, 10^6, 10^8$.)

The tail. By the Azuma–Hoeffding inequality, for $J' > J$,

$$\mathbb{P}(|M_{J'} - M_J| \ge u) \le 2\exp\Big(-\frac{u^2}{2\sum_{j>J} a_j^2}\Big), \qquad \sum_{j>J} a_j^2 \le C'^2 \sum_{\ell\ge z} \frac{(\log\ell)^{2B_0}}{\ell^2} \ll \frac{(\log z)^{2B_0}}{z}.$$

As $J'\to\infty$, $M_{J'} - M_J \to \mathcal{Y} - M_J$ in probability, so $\mathbb{P}(|\mathcal{Y}-M_J|>u) \le \liminf_{J'} \mathbb{P}(|M_{J'}-M_J|\ge u)$, and the lemma follows with $B_1 = 2B_0$. □

Proof of Theorem 4. (a) Fix $\varepsilon\in(0,1)$ and let t be large. Take $\log z = (1-\varepsilon)\sqrt{t/\kappa_H}$ and $u = \varepsilon t$ in Lemma 6.5. Then $(\kappa_H+\rho_H(z))(\log z)^2 = (1-\varepsilon)^2(1+o(1))t \le (1-\varepsilon)t$, so

$$\nu\big((c_0+t,\infty)\big) + \nu\big((-\infty, c_0-t)\big) = \mathbb{P}(|\mathcal{Y}|>t) \le 2\exp\Big(-c\varepsilon^2 t^2 \frac{z}{(\log z)^{B_1}}\Big) = \exp\big(-z^{1+o(1)}\big),$$

because $t \asymp (\log z)^2$. That is $\exp(-\exp((1-\varepsilon)(1+o(1))\sqrt{t/\kappa_H}))$, and ε is arbitrary. With Corollary 6.1 this proves (a).

(b) Finiteness of all exponential moments follows from (a). Let $\delta\in(0,1)$ and θ be large, and write $\mathcal{Y} = Z - c_0$. *Lower bound:* with $t = (1-\delta)\kappa_H(\log\theta)^2$, (a) gives $\mathbb{P}(\mathcal{Y}>t) \ge \exp(-\theta^{1-\delta/4})$ for large θ, so $\mathbb{E}e^{\theta\mathcal{Y}} \ge e^{\theta t}\mathbb{P}(\mathcal{Y}>t)$ gives $\log\mathbb{E}e^{\theta\mathcal{Y}} \ge (1-\delta)\kappa_H\theta(\log\theta)^2 - \theta^{1-\delta/4}$. *Upper bound:* $\mathbb{E}e^{\theta\mathcal{Y}} = \int_{\mathbb{R}} \theta e^{\theta t}\mathbb{P}(\mathcal{Y}>t)\,dt$. Split at $t_1 = (1+\delta)\kappa_H(\log\theta)^2$. The part $t<t_1$ is at most $e^{\theta t_1}$. For $t\ge t_1$ put $s = \sqrt{t/\kappa_H} \ge (1+\delta)^{1/2}\log\theta$. By (a), $\mathbb{P}(\mathcal{Y}>t) \le \exp(-e^{(1-\delta')s})$ for large t, with $\delta'>0$ so small that $(1-\delta')(1+\delta)^{1/2} > 1+\delta/8$. Then $e^{(1-\delta')s} \ge 2\theta\kappa_H s^2 = 2\theta t$ for θ large, so the integrand is at most $\theta e^{-\theta t}$, and this part is at most $e^{-\theta t_1}\le 1$. Hence $\log\mathbb{E}e^{\theta\mathcal{Y}} \le (1+\delta)\kappa_H\theta(\log\theta)^2 + 1$. The same holds for $-\mathcal{Y}$, since (a) is symmetric.

(c) By Theorem 1(a) the empirical distributions of X_p, $p\le x$, converge weakly to ν. By the portmanteau theorem, $\nu((c_0+t,\infty)) \le \underline{\delta}_+(t) \le \overline{\delta}_+(t) \le \nu([c_0+t,\infty)) \le \nu((c_0+t-1,\infty))$, and similarly for δ_-. Now apply (a) to t and $t-1$. □

Corollary 6.6 (Uniform lower bound for primes)**.** *For every $\varepsilon>0$ there are $t_0(\varepsilon)$ and $X_0(\varepsilon)$ such that, for $X\ge X_0(\varepsilon)$ and $t_0(\varepsilon)\le t\le(\kappa_H-\varepsilon)(\log\log X)^2$,*

$$\#\{p\in(X,2X]:\ \pm(X_p - c_0) > t\} \ \ge\ \pi(X)\exp\Big(-\exp\big((1+\varepsilon)\sqrt{t/\kappa_H}\big)\Big).$$

Proof. Run the proof of Theorem 3 with a given K in place of the maximal one: $Q = Q_K$, $Y = Q(\log X)^{16}$ and $T = \max(QY^2, \exp(\sqrt{\log 2X}))$. This is allowed for $5\le K\le K_{\max}(X)$, where $K_{\max}$ is the value chosen there, since then $T\le T_0$. Every error term in that proof is

uniform in K in this range. The function $\eta(K)$ of Lemma 5.2 may be replaced by $\sup_{K'\ge K}\eta(K')$, and the deterministic bound of Lemma 5.3 depends only on K. In the exceptional case, $\ell_1\to\infty$ as $X\to\infty$, because $\log\log T\ge(1+o(1))\log K$ and $\log\log T\ge\frac12\log\log X$. Hence, for $K\ge K_1(\varepsilon)$ and $X\ge X_0(\varepsilon)$, at least $\frac12 X/(\varphi(Q_K)\log X)$ primes $p\in(X,2X]$ satisfy $\pm(X_p-c_0)\ge(\kappa_H-\varepsilon/4)(\log K)^2-1$. Given t, we may assume $\varepsilon\le\varepsilon_0$ is small, since the statement weakens as ε grows. Take K minimal with $(\kappa_H-\varepsilon/4)(\log K)^2-1>t$. Then $\log K\le\sqrt{(t+1)/(\kappa_H-\varepsilon/4)}+o(1)\le(1+\varepsilon/2)\sqrt{t/\kappa_H}$ for $t\ge t_0(\varepsilon)$, and $K\le K_{\max}(X)$ by the upper limit on t. Finally $\varphi(Q_K)\le e^{(4+o(1))K}$, and $4K\le\exp((1+\varepsilon)\sqrt{t/\kappa_H})$ for $t\ge t_0(\varepsilon)$. □

Remark 6.7. For fixed t this is the lower half of Theorem 4(c). At the top of the range, $t\asymp(\log\log X)^2$, it is Theorem 3 again (up to ε). So the tail of ν is visible in the primes, from below, uniformly up to the extreme scale. A matching uniform upper bound for primes is open; for fixed t it is Theorem 4(c).

The martingale argument also gives all moments over the primes, which extends Theorem 1(a) beyond quadratically bounded test functions.

Corollary 6.8 (All moments)**.** *For every continuous $f:\mathbb{R}\to\mathbb{R}$ of polynomial growth,*

$$\frac{1}{\pi(x)}\sum_{p\le x}f(X_p)\to\int f\,d\nu\qquad(x\to\infty).$$

In particular $\frac{1}{\pi(x)}\sum_{p\le x}(X_p-c_0)^m\to\mathbb{E}[(Z-c_0)^m]$ for every integer $m\ge1$.

Proof. Uniform sub-Gaussian bound. For $Y\ge3$ the proof of Lemma 6.5 applies verbatim to $\frac{4}{\pi}R_Y$ in place of $\mathcal{Y}$, with the weights $w_d(Y)\mathbf{1}_{d<Y}\in[0,\pi^2/8]$ in place of $\pi^2/8$. The martingale is then finite, and every bound used only that the weights lie in $[0,\pi^2/8]$. Take $J=0$: then $M_0=\mathbb{E}[R_Y]=0$, and Azuma–Hoeffding with $\sum_j a_j^2<\infty$ gives $\mathbb{P}(|R_Y|>u)\le2e^{-cu^2}$, with $c>0$ absolute and the bound uniform in $Y\ge3$. Hence $\sup_{Y\ge3}\mathbb{E}|R_Y|^r<\infty$ for every $r\ge1$.

Bounded moments over primes. Fix $m\ge1$, let X be large, $K_2=\sqrt{2X}(\log 2X)^4$ and $Y=X^{1/(5m)}$. For $p\in\mathcal{P}_X$, Theorem B(ii) gives $X_p-c_0=\frac{4}{\pi}(R_Y(p)+V(p))+O(1/\log X)$ with $V=R_{K_2}-R_Y$, as in Lemma 3.2.

- *The part V.* By Corollary 4.3(b), $|R_K(n)|\le\frac1\pi\sum_{k<K}\frac{\log k+K_0}{k}\ll(\log K)^2$ for every integer n, so $|V(n)|\ll(\log X)^2$. By Lemmas 9.3(b) and 9.4, $\sum_{X<n\le2X}V(n)^2\ll X(\log X)^{14}/Y$. Hence

$$\sum_{p\in\mathcal{P}_X}|V(p)|^{2m}\ll(\log X)^{4m-4}\cdot\frac{X(\log X)^{14}}{Y}=o(\#\mathcal{P}_X).$$

- *The part R_Y.* Expand R_Y^{2m} as a sum over $(d_1,\dots,d_{2m})$, $3\le d_i<Y$, of $\prod_i w_{d_i}(Y)G_{d_i}/d_i^2$. Each product is a function f of p mod q, $q=\operatorname{lcm}(d_i)\le Y^{2m}\le X^{2/5}$. So

$$\sum_{p\in\mathcal{P}_X}f(p)=\#\mathcal{P}_X\,\mathbb{E}[f]+\sum_{(a,q)=1}f(a)\,E(X;q,a),$$

  where $E(X;q,a)$ is the error term for primes in $(X,2X]$ that are $\equiv a\pmod q$, relative to $\#\mathcal{P}_X/\varphi(q)$. Replacing $\#\mathcal{P}_X$ by $\mathrm{li}(2X)-\mathrm{li}(X)$ changes $\sum_{q\le X^{2/5}}\tau(q)^{4m}\max_a|E|$ by only $O_m(X\exp(-c\sqrt{\log X}))$, and trivially $\max_a|E(X;q,a)|\ll X/\varphi(q)$. By Lemma 9.2(b), $\|f\|_\infty\le\prod_i\frac{\pi^2}{8}\cdot\frac{\tau(d_i)d_i(\log d_i+2)}{\pi d_i^2}$, so the error for one tuple is at most $\varphi(q)\|f\|_\infty\max_a|E(X;q,a)|\le(C\log X)^{2m}\prod_i\tau(d_i)\max_a|E(X;q,a)|$, because $q\le\prod_i d_i$. A given q arises from at most $\tau(q)^{2m}$ tuples, each with

$\prod \tau(d_i) \le \tau(q)^{2m}$. The Bombieri–Vinogradov theorem with divisor weights (by Cauchy–Schwarz and the trivial bound $\max_a |E(X;q,a)| \ll X/\varphi(q)$) gives $\sum_{q\le X^{2/5}} \tau(q)^{4m} \max_a |E(X;q,a)| \ll_{m,A} X(\log X)^{-A}$. Hence

$$\sum_{p\in\mathcal{P}_X} R_Y(p)^{2m} = \#\mathcal{P}_X\, \mathbb{E}[R_Y^{2m}] + O_m\big(X(\log X)^{-2}\big) \ll_m \#\mathcal{P}_X,$$

by the uniform sub-Gaussian bound.

By Minkowski's inequality, $\sup_X \frac{1}{\#\mathcal{P}_X} \sum_{p\in\mathcal{P}_X} |X_p - c_0|^{2m} < \infty$, and therefore also over $p \le x$.

Conclusion. Since m is arbitrary, the empirical distributions of X_p, $p \le x$, have uniformly bounded moments of every order. Together with their weak convergence to ν (Theorem 1(a)), this gives uniform integrability of $|X_p|^r$ for every r, and the claim follows. □

Remark 6.9. The data match the moments of ν beyond the variance. For the 302302 primes in $(10^6, 5.5\cdot 10^6]$, $X_p - c_0$ has skewness -0.0006 and excess kurtosis -0.424. For $4\cdot 10^5$ Haar samples of $\frac{4}{\pi}R_{8001}$ the values are -0.0005 and -0.417. (Standard errors are about 0.004 for the skewness and 0.006 for the kurtosis.) The negative excess kurtosis says that at the scale of σ the law ν is flatter than a Gaussian. That is a separate fact from the far-tail asymptotics of Theorem 4(a), which concern only $t \to \infty$.

Remark 6.10 (A sharp form of the conjecture). Theorem 4 suggests the precise size of the extreme values. If the X_p behaved like independent samples of ν, then by (a) the probability that $X_p - c_0 > (1+\varepsilon)\kappa_H(\log\log p)^2$ would be $\exp(-(\log p)^{(1+\varepsilon)^{1/2}+o(1)})$. That is summable over p. With $1-\varepsilon$ in place of $1+\varepsilon$ the sum diverges. This leads to the following conjecture.

Conjecture 5. *We have*

$$\limsup_{p\to\infty} \frac{X_p - c_0}{(\log\log p)^2} = \kappa_H, \qquad \liminf_{p\to\infty} \frac{X_p - c_0}{(\log\log p)^2} = -\kappa_H.$$

Remark 6.11. By Theorem 3, the inequalities $\limsup \ge \kappa_H$ and $\liminf \le -\kappa_H$ hold unconditionally. The open part is the upper bound $|X_p - c_0| \le (\kappa_H + o(1))(\log\log p)^2$. It is the analogue, for $S(p)$, of the conjecture of Montgomery and Vaughan on the extreme values of $L(1,\chi)$ (H. L. Montgomery and R. C. Vaughan, *Extreme values of Dirichlet L-functions at* 1, in: Number Theory in Progress, de Gruyter, 1999, 1039–1052). It has no bearing on Conjectures 1.1, 3.1 and 3.2 of [6] ($S(p) > 0$, $> 2p$, $> 3p$). Their eventual forms follow from the Bourgain–Garaev bound for short Kloosterman sums [1], and what remains open there is the intermediate range of p, which a statement with $o(1)$ cannot reach. The data are consistent with it: for $10^3 < p \le 5.5\cdot 10^6$ the ratio $(X_p - c_0)/(\log\log p)^2$ lies in $[-0.354, 0.341]$, and it is -0.306 for the extremal prime $p = 9166207049$ of [1, Table 4]. All these values are below $\kappa_H = 0.361$ in absolute value. Since $\log\log p$ grows so slowly, this is weak evidence.

7. Open questions

(1) The upper half of Conjecture 5: is $|X_p - c_0| \le (\kappa_H + o(1))(\log\log p)^2$ for every prime p?
(2) Is the support of ν all of $\mathbb{R}$? Is ν absolutely continuous, with a smooth density? The character expansion of Theorem 2 writes R_∞ as a series in characters of independent local components, but its terms are not independent.
(3) Give a rate of convergence in Theorem 1, and make $X_0(\varepsilon)$ in Theorem 3 explicit. The second needs explicit constants in Gallagher's theorem.
(4) The conjectures $S(p) > 0$, $S(p) > 2p$, $S(p) > 3p$ of [6] remain open in the range between computation ($p \le 6\cdot 10^6$) and the non-explicit threshold that follows from the Bourgain–Garaev bound. Closing it would need explicit bounds for short Kloosterman sums far beyond the current ones.

Use of AI

This work was done with substantial assistance from Claude, an AI system developed by Anthropic, which under the direction of the author found and wrote most of the proofs and the code. The proofs have been checked by computer and by automated referee runs of an AI system, which are not independent of the system that helped produce them. The paper has not yet been checked by a human expert, and the author takes responsibility for any errors. The results rely on the main paper [1].

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Email address: samsonmfraser@gmail.com